Linear A Texts & Inscriptions in phonetic transcription

& Commentary

Introduction

inaugural date: 30 November 2000; last update: 8 April 2024 30 minutes after the solor eclipse totality

For all updates, arranged by date of posting, see the file "Updates."

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1. Fonts

The following fonts are available for Macintosh OS 9 (NOT OS X) (courtesy Jean-Pierre Olivier):
 Phaistos Disc - "Phaistos 12.hqx", "Phaistos 48.hqx"
 Cretan Hieroglyphic - "Mobile 18.hqx" (for clay Hieroglyphic texts), "Malia-thick 18.hqx" (for Hieroglyphic sealstones)
 Linear A - "Knossos 18.hqx"
 Linear B - "Mycenae 12.hqx", "Mycenae 18.hqx"
 Linear B Ideograms "ID. B 18.hqx"

The following fonts are now available (7 Sep 08) for Macintosh OS X (courtesy Jean-Pierre Olivier):

 Cretan Hieroglyphic for MAC OS X
Malia-Maigre for inscriptions
Malia-Gros for sealstones and seal impressions

For **Windows**, David Willem Borgdorff has designed a Linear A font: "LA.ttf.hqx"
The fonts have been stored in binhex mode; after you download them (ignore all warnings), they must be "unstuffed" (for a **free** "Unstuffit" program; quit all programs, install the fonts in your Macintosh/Windows Harddrive/Systems/Fonts folder, and then restart your word-processing package).

2. Corpora and Phonetic Transcriptions

The texts presented here are based on the texts in GORILA (below) and a transnumeration and phonetic normalization finished 22 March 1994 by John G. Younger; Jean-Pierre Olivier checked this document against GORILA vols. I-V and a ms. of VI. It was then put in tabular form in January-February 1997. Since then, there have been continual updates. From 1994/7 to April 2024, this online edition of Linear A was housed on a sub-server of Kansas University (people.ku.edu/~jyounger/ etc.). KU then removed support of this server at the beginning of February 2024 and Younger began remediating this folder into MS Word docx and PDF files in preparation for storing them on a publicly accessible website.

GORILA = Louis **G**odart and Jean-Pierre **O**livier, **R**ecueil des **i**nscriptions en **L**inéaire **A**. Études Crétoises 21, vols. 1-5. (Paris: Librairie Orientaliste Paul Geuthner, 1976-1985, now available online, along with the rest of the Études Crétoises volumes).
 Volume 1: Tablettes éditées avant 1970 (EtCret 21:1; Paris: Librairie Orientaliste Paul Geuthner 1976) ISBN X16534
 Volume 2: Nodules, scellés et rondelles édités avant 1970 (EtCret 21:2; Paris: Librairie Orientaliste Paul Geuthner 1976) ISBN X16435
 Volume 3: Tablettes, nodules et rondelles édités en 1975 et 1976 (EtCret 21:3; Paris: Librairie Orientaliste Paul Geuthner 1976) ISBN X16436
 Volume 4: Autres documents (EtCret 21:4; Paris: Librairie Orientaliste Paul Geuthner 1982) ISBN X16437
 Volume 5: Addenda, corigenda, concordances, index et planches des signes (EtCret 21:4; Paris: Libraire Orientaliste Paul Geuthner, Paris 1985) ISBN X16433

The phonetic transcriptions use Linear B values for Linear A signs assumed to be the same, according to the following principles. **Also see below, "Phonetic values of the signs."**

Words of 3 or more syllables appearing exactly the same in both Linear A and Linear B

Godart 1984
A-TI-KA (ZA Wc 2.a1-2; KN V 831.4)
DA-I-PI-TA (ZA 8.5, 10a.4-5; KN B 799.1)
I-JA-TE (PH Zb 4; PY Eq 146.9)
I-TA-JA (HT 28a.6; KN Ap 769.2, Xe 537.2)
KI-DA-RO (HT 47a.4, 117a.9, 122a.2-3?; KN E 842.3)
PA-I-TO (HT 97a.3, 120.6; KN 59 occurrences)
SE-TO-I-JA (PR Za 1b; KN 22 occurrences)
SU-KI-RI-TA (PH Wa 32; KN 9 occurrences), SU-KI-RI-TE-I-JA (HT Zb 158b)
possibly A-RA-KO (KO? Zf 2; KN 5 occurrences)

Additional A-TI-KA (ZA Wc 2.a1-2; KN V 831.4)
PA-RA-NE (HT 115a.4, b.1; KN Vc 7616)

Words of 3 or more syllables in Linear A that are very similar to words in Linear B

A-KA-RU (HT 2.1, 86a.1); cf. *a-ka-re-u* (KN B 416)
A-RA-NA-RE (HT 1.4); cf. *a-ra-na-ro* (KN As 1516.11)
A-RE-SA-NA (THE Zb 2); cf. *a-re-sa-ni-e* (PY An 724.2)
A-SA-RA_2 (HT 89.1); cf. *a-sa-ro* (KN As 40.4)
A-SU-JA (HT 11a.3-4); cf. *a-si-wi-ja* (PY Fr 1206)
A-TA-RE (ZA 8.1); cf. *a-ta-ro* (PY An 35.5)
A-TI-RU (ZA 4a.3); cf. *a-ti-ro* (KN Dv 1272B)
DI-DE-RU (HT 86a.3, 95a.4, b.4); cf. *di-de-ro* (KN Dv 1504B)

DA-MI-NU (HT 117a.8); cf. *da-mi-ni-ja* (PY Aa 96, Ad 697) and *da-mi-ni-jo* (common on KN D-tablets and elsewhere)
DU-PU3-RE (KO Za 1b); cf. *du-pu2-ra-zo* (KN Da 1173)
KA-SA-RU (HT 10b.3); cf. *ka-sa-ro* (KN C 912B)
KI-RI-TA2 (HT 114a.1, 121.1); cf. *ki-ri-ta* (KN G 820.1, Ld 785.1)
KU-KU-DA-RA (HT 117a.7); cf. *ku-ka-da-ro* (KN Uf 836.b)
KU-PA3-NU (HT 1, etc.); cf. *ku-pa3-no* (KN Df 1219B)
KU-PA3-NA-TU (HT 47a.1-2, 119.3); cf. *ku-pa-nu-we-to* (KN As 1517.8)
KU-RU-KU (HT 87.4); cf. *ku-ru-ka* (KN Vc 5510)
MA-SI-DU (HT 43.1-2); cf. *ma-si-dwo* (KN Fh 360B) MI+JA+RU (several); cf. *mi-ja-ro* (KN Ln 1568.1)
PA-JA-RE (several); cf. *pa-ja-ro* (KN As 1519.6)
QA-QA-RU (several); cf. *qa-qa-ro* (KN As 604.3)
QA-RA2-WA (HT 86a.3); cf. *qa-ra2-wo* (KN Ce 50.1a)
SA-MA-RO (HT 88.5-6); cf. *sa-ma-ra* (PY Jn 829.15, etc.,]*sa-ma-ru*[, KN V 655.1, and *sa-ma-ri-jo*, KN Da 1147, Np 857)
SI-KI-RA (HT 8a.4); cf. *si-ki-ro* (KN U 8210.1)
SI-MI-TA (HT 96a.2-3); cf. *si-mi-te-u* (KN Am 827.1)
TA-NA-TI (HT 7a.4, 10b.4, 98a.2); cf.]*ta-na-ti* (KN Uf <311>.2)
TE-JA-RE (HT 117a.5); cf. *te-ja-ro* (KN V 479 v.3, X 5525.1, 8661B)
WA-DU-NA (HT 6b.1-2, 85b.4-5); cf. *wa-du-na* (KN V 503.3)
WI-RA-RE-MI-TE (ZA 9.6); cf. *we-ru-ma-ta* (PY Ub 1318.4)

3. Conventions (definition of "inscription"; bibliographical & epigraphical conventions)

a. **Definition of "inscription"**
By epigraphic convention, an "inscription" is defined as two or more signs in conjunction (e.g., a word). A single sign may be an abbreviation of a word (e.g., Linear B *o* for *o-pe-ro*, "owed", "deficit"), or, commonly in Linear A, a transaction sign, like AB sign *42 A ("to" NAME?) and TE ("from"? NAME).
Thus, pot marks and other individual and solitary marks "must be clearly distinguished from writing in the true sense of the word, and their forms generallly have only a fortuitous resemblance to alphabetic or linear writing signs" (DOCs^2 p. 29; cf. further remarks that disassociate such signs from lexical meaning on pp. 40 and 63).
Thus, this document does not list pot marks, mason's marks, sings on sealstones, and other single & solitary signs that resemble Hieroglyphic, Linear A, or Linear B signs.

b. **Bibliography** from 1980 (with seminal works prior)

Abbreviations

- *DOCs^2* = M. Ventris and J. Chadwick, *Documents in Mycenaean Greek* (ed. 1), Cambridge: Cambridge University Press 1956; second edition by J. Chadwick, Cambridge University Press 1973.

- CMS = *Corpus der minoischen und mykenischen Siegel* (Berlin: Gebr. Mann Verlag 1964-2000; and Mainz: Philipp von Zabern 2002-present).
- GORILA: see section 3, above.

c. **Epigraphical Conventions**

SIGN[= text broken off at left
]SIGN = text broken off at right
SIGN = the reading of SIGN is doubtful
• WORD / SIGN • (e.g., • TE • or • KI-RO •) = Transaction Sign or Word)
WORD • WORD (i.e., • flanked by spaces) = word-divider
-• or •- or -•- = unidentified sign
[[SIGN]] = erased but legible sign
<TEXT> = TEXT once extant, now lost and supplied

d. **Types of Supports** (the objects that receive writing)

no specialized designation = tablets (like Linear B "page" tablets)
Wa = nodules/*noduli*
Wb = sealings
Wc = roundels
Za = stone vessels
Zb = clay vessels
Zc = inked inscriptions
Zd = graffiti
Ze = architecture
Zf = metal objects
Zg = stone objects
rare document shapes carried over from Hieroglyphic (but given no specialized designation in GORILA)
- bars, 3-sided (like the common Hiero Class Hg) -- MA 1, 2
- bars, 4-sided (like Hiero Class Hh) -- MA 10; and PK 3
- labels (derived from Hiero crescents, Class Ha, or medallions, Class He) -- PH 10? & 24; and ZA Zg 35
- lames (short, thin, flat, horizontal tablets; cf. Hiero Class Hf, lames [e.g., SY Hf 01]) -- PH 12-17, 22, 16; PYR 2?; and KN 22

4. Basic Statistics

There are over 1500 Linear A documents with a total occurrence of about 7500 signs (augmenting Schoep 2002, 38); if there are 4002 characters (font Times, pitch 12, no spaces) on a 8 1/2 x 11 inch sheet of paper with 1 inch margins, all extant Linear A would take up 1.84 pages.

Contrast Hieroglyphic: some 360 documents [including seals and the 23 documents from Petras] with fewer than a total occurrence of 1000 signs.

Contrast Linear B: some 4600+ documents with 57,398 signs -- 14.34 pages of text.
About 2,788 people seem to be recorded in the Linear A tablets (Schoep 2002, 185).

5. My Goals for Establishing This Document

I have three major goals for this document.

1. to make available the results of GORILA to the wider public (with permission of Louis Godart and Jean-Pierre Olivier). This document is not a substitute for GORILA since it "normalizes" and phonetically transcribes GORILA's layouts and drawings of the Linear A documents.

2. to work publicly with colleagues on the Minoan administrative process: how commodities were assessed, collected, and distributed.

3. to tease, from an understanding of the administrative process, the administrative vocabulary (e.g., the terms for "total" and for "deficit") and its grammar and syntax.

I am NOT interested in producing a decipherment of Linear A -- please see "Decipherments," below.

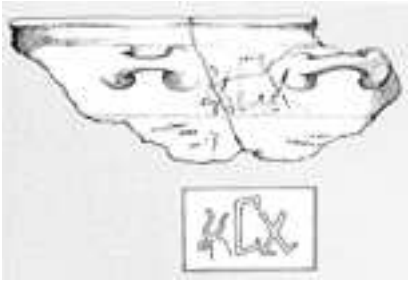
6. What Is Known about Linear A

6a. Chronology

Hieroglyphic is probably the first script to appear; seals dating from primarily MM I contexts (MM IA or IB) include CMS II.1 nos. 391, 393, and 394 from Archanes Phournoi (the signs on these seals are often referred to as comprising "the Arkhanes script"; Schoep 2002, 23: early form of Linear A?).

Hieroglyphic documents come from four main deposits in Crete: Malia, Quartier Mu (MM II late); Petras (MM IIB); Malia Palace (MM [IIB-]III); and the Knossos Hieroglyphic Deposit (an assemblage of material from the end of the Long Corridor in the West Wing and surrounding area). Since the sealstones that impressed the KN sealings were all hard stone seals, the impressed sealings at least should date to MM IIB-III. No Hieroglyphic document can be dated later than MM III.

Linear A documents appear either contemporary with or soon after the first appearance of Hieroglyphic writing. The earliest document may be ARKH Zc 8, a painted larnax rim which the excavators first dated to an EM II-MM IA context (1975) but later assigned it to the "Old Palace

period" (1997; i.e., MM IB at the earliest).  The first sign looks Hieroglyphic, like the bull head H012 ; the other two signs are definitely Linear A: TA-JE. If the inscription is indeed a mix of scripts, it may well be very early.

The other earliest documents date to MM IIA (KN 40 from Knossos, South House, carrying a badly legible fraction) or MM II (ARKH Zf 9; PH 6-19, 22, 24-28, 30 [Haghia Photini], Wb 33-36, Wc 37-41, 43, 44, 46, 52, 55, Wg 45, and Wy 42; and SAM Wa 1).

Hieroglyphic was therefore probably invented first, in MM IA and appears first on seals from Archanes and Ayia Triada; Linear A follows immediately in MM IB, or soon after, in MM II, and appears first on documents primarily from Phaistos. From then until MM III, Hieroglyphic and Linear A were being written contemporaneously, with Hieroglyphic documents at Malia Quartier Mu (MM II) and Malia Palace (MM III, and Knossos Palace?) and with Linear A documents at Phaistos (MM II), Malia Palace (MM III), and Knossos Palace (MM IIIB). From this evidence, it is possible that Hieroglyphic originated in central Crete first (or possibly at Malia), in MM IA and Linear A originated at Phaistos slightly later in MM II.

Although the two scripts share several signs, which may have similar phonetic values, it is not clear why two such different scripts should have developed more or less contemporaneously unless they represent two different administrative practices and/or two different languages or dialects (Schoep 2002, 22-23).

A few Linear A inscriptions come from Final Palatial contexts. Three occur on objects that probably were made earlier but were found in later contexts: ARM sealstone, KH 94 and Wc 2117 & 2118 (all from fills), and the MM III pithos KN Zb 35 (LM II context). Three others, however, were probably all written after the LM Ib destructions: the LM II pithoid jar KN Zb 40, the inscribed block KN Ze 16 on the Kephala tholos (LM II?), and the painted inscription on PO Zc 1, a terracotta statuette from Poros (LM III A:1-2 by form and context). It is therefore possible that Linear A survived the LM Ib destructions, though barely.

In the Neopalatial period, and probably as early as LM IA, Linear A was developing in such a way as to produce Cypro-Minoan and, eventually, Linear B. The earliest CM document, the "Grand" tablet from Enkomi comes from a 16th century context. The earliest Linear B documents seem to be those from the Room of the Chariot Tablets at Knossos (LM II or possibly LM III A1). This would be the time T.G. Palaima would identify for the creation of Linear B, when, on the Mainland, major Mycenaean centers had major administrative buildings on their heights, the larger tholoi were being constructed, and trade was expanding. According to Palaima, Linear B was created probably at a single moment, either "from above" (a directive) or

"within the close confines of palatially oriented bureaucracies which then sanctioned the invention and continued to apply it to narrow record-keeping tasks" (1988: 341).

Some Hieroglyphic signs, not Linear A signs, are the prototypes for Linear B signs (e.g., CHIC sign 40 for Linear B *ro2* and 78 for Linear B *do*); it is therefore possible that Linear B was developing earlier than LM/LH II and incorporating more than one source.

Other linear scripts may have similarly developed from Linear A farther east: see the inscription from Lachish (Finkelberg 1996).

Character of the Script and Documents

6b. The Script

In comparison to Cretan Hieroglyphic, Linear A is fairly neat, written (almost?) always from left to right (a few examples of sinistroverse or boustrophedon writing: IO Za 9 & 11; KN Za 19; PL Zf 1; VRY Za 1) in more or less straight rows from top to bottom of the clay document; occasionally the tablets are partially ruled. The signgroups are separated by a dot or short stroke. The longest texts are HT 93a & b, HT 117a & b.

Linear A contains more than 90 signs in regular use and a host of logograms, many of which are ligatured with syllabograms and/or fractions; About 80% of these do not appear in Linear B. While many of the regular signs are also found in Linear B, some signs are unique to A (e.g., A *301 and following), while some signs found in Linear B are not found in Linear A (e.g., AB 12, 14-15, 18-19, 25, 32-33, 35-36, 42-43, 48, 52, 62-64, 68, 71-72, 75, 83-85).

There is probably much Minoan retained in Linear B. Signs AB *22 PA_3 and *56 (unknown phonetic value, logogram B *107 CAPrid) were retained in Linear B for representing foreign (Minoan) sounds, and there have been attempts to identify Minoan names in Linear B (Billigmeier 1969, Firth 1993). Some Minoan words are retained in Linear B and even become Hellenized, like Linear A MA+RU becoming Linear B's logogram *145 LANA and Greek μαλλός, mallós.

It is well-known that Linear A uses three main vowels, A, I, U; Linear B adds e- and o-series, and complex phonemes (e.g., dwo, two).

The complex phoneme nwa, however, is attested in Linear A on SY Za 4 and may be implied in the word KU-PU_3-NA-TU; compare Linear B *ku-pa-nu-we-to*, a man's name on KN As 1517.8 (thanks to Gretchen Leonhardt for bringing this to my attention).

Meléna 2014, 115: "The use of graphic glides after /i/ and /u/ when followed by a vowel" (e.g., -ija, -pwa) is probably coming from Linear A (e.g., Linear B *se-to-**i-ja***, many KN tablets, duplicating Linear A SE-TO-**I-JA**, PR Za 1).

Compare

Linear B	Linear A
*76 *ra*$_2$ /rja/	PA-SA-RI-JA (HT 24a.4), KU-PA$_3$-RI-JA (HT 24a.1)
a-pu-wa (TH Fq 229.3)	A-SU-PU-WA (ARKH 2.5-6)
*86 *d*^{w}*a*$^{??}$	DA-WA-[•]-DU-WA-TO (KN Za 10a-b)
*19 *ru*$_2^{??}$ /rju/	*309a+RI-JU (logogram on TY 2.3, .6)
*92 *sja*$^{??}$ (PY Ma 397[+]1048.1: *a-•-ta*$_2$ = *a-*92-ta*$_2$)	A-SI-JA-KA (HT 28.1, b.1-2) and common ending -SI-JA-SE

6c. The Documents

Linear A was written on a variety of supports, unlike Linear B: stone offering tables, gold and silver hair pins, and pots (inked and inscribed). The clay documents consist of tablets, roundels, and sealings (one-hole, two-hole, and flat-based). Roundels relate to the "conveyance of a commodity, either within the central administration or between the central administration and an external party" (Schoep 2002, 122) with the roundel being the record of this transaction that stays within the central administration as the commodity moves out of the transacting bureau (see Hallager 1996a). Two-hole sealings probably dangled from commodities brought into the center, one-hole sealings apparently dangled from papyrus/parchment documents, and flat-based sealings were pressed against the twine that secured papyrus/parchment documents (see CMS II6-II8 for photographs of the imprints that survive on the underside of flat-based sealings). These papyrus/parchment documents, presumably carrying inked texts, were probably more important than the clay tablets and roundels that have survived.

Most of the clay documents are page-shaped tablets, usually smaller than Linear B tablets; there are a few bars taken over from Hieroglyphic, but this shape is discontinued after MM III. While there are a few oblong tablets (e.g., from Malia), there are no palm-leaf tablets like those of Linear B.

Those tablets that are written on both sides (opisthographic) are simply turned over (along the vertical axis) from the front (recto) to the back (verso), like a modern book page, unlike the Cypro-Minoan tablets from Enkomi, which turn over along the horizontal axis like a notepad. Schoep 2002, 73, says the opisthographic "Linear A tablets always rotate around their horizontal axis"; this probably just a simple mistake.

As Linear A was written on a variety of supports, so too we find the documents in a wide variety of settings: private houses at Khania and at Ano Zakros, decentralized bureaux (the Casa del Lebete at Haghia Triada, the Northeast House at Knossos, and Zakros Hogarth's House A), and central administrative areas (the Villa Reale at Haghia Triada and the palaces at Knossos, Malia, and Zakros). Most of the documents come from "living" archives (e.g., Zakros palace, Haghia Triada's Magazines 5-7, HT 24 recording wool and found with loomweights) or "discard" archives (Knossos Temple Repositories and West Magazines, Malia Palace). Often the discard archives are found fallen from an upper storey (e.g., Knossos West Magazines and Zakros Hogarth's House A), and some were stored in chests of wood (e.g., Khania; bronze hinges from

Knossos Temple Repositories, Zakros rooms XVI, XXVIII, Tylissos House A room 5) or clay or mudbrick (Zakros House A, Khania house I, Palaikastro Building V) (Schoep 2002, 25-26).

Almost all tablets are palimpsests (i.e., they have been written on, erased, and written on again). "The presence of palimpsests suggests that the information originally held on the tablet was discarded, perhaps because it had been copied onto another document and/or updated" (Schoep 2002 79).

Almost all documents consist of lists, with headings, of one or two word entries, each followed by a logogram followed by a number and/or fraction. Once, possibly twice, the logogram is followed by a word and then by a number; this "constantly happens in Linear B" (Hooker 1975). On HT 88.2, the ideogram FIC (for "figs") is followed by the word KI-KI-NA, which may designate a type of fig (black vs. green) or an aspect (fresh vs. dried). On KN Za 19, the ideogram *188 or TALENT is followed by MI-NA, which may be the name of the weight, the "mina" (cf. ZA 21a.7).

Most words seem to be names (person- , place-) (Hooker 1975); when in doubt as to whether a name is an anthroponymn or toponymn (which is almost always), I tend to write "NAME." A few words may be verbs or transaction words (see below).

We should imagine the several stages in recording information in the Minoan bureaucracies:

1. information on individual transactions
2. compiling these individual transactions, say by region or person
3. summarizing these individual transactions, say by commodity
4. making permanent or final records
5. discarding the preliminary documents

The types of transactions recorded should be fairly limited:

- inventories -- of storerooms (food, raw materials, finished products), stables, workshops, lands belonging to the administration, their goods & flocks. "Inventories form the basis of target records" (Schoep 2002, 90-91).
- assessments -- the central administration telling outlying regions what they should produce; a kind of taxation
- collections, contributions -- <u>inward</u> movement of goods and contributions to the central administration (in response to assessments, taxes, levies), deliveries, sacrifices, etc.; commodities recorded as in-coming from the outlying regions, either actually arrived (KU-RO, total) or in arrears (deficits, KI-RO).
- allocations, disbursements, distributions -- recording the <u>outward</u> movement of agricultural commodities, goods, animals and raw materials for various purposes (finishing, pay, rations, trade, etc.); commodities given back out to people or places as rations or payments for services rendered. Mixed commodities and combined documents (Schoep's type IB & V) with VIR and "the Aegean triad" (FIC, GRA, VIN) probably record allocations (e.g., HT 100). See PE 1, KH 7, HT 27, HT 39.

The scribes identified at Haghia Triada did not specialize in recording certain commodities (except perhaps Hand 9 recording people, Schoep 2002, 167 n. 264, 199 n. 110); and it is not possible usually to assign documents to a single series since most documents record, individually, a variety of commodities. The texts also differ from site to site; for instance, Haghia Triada and Zakros tablets contain more lexical information but fewer logograms than do the documents from Khania. We must imagine that practices differed from site to site and/or that we have texts from the different stages of processing information.

Packard 1974 classifies the Haghia Triada documents into 9 categories based on subject matter and format:

A series: words followed by 1
B series: logogram 100/102 VIR & ligatures
C series: agricultural products
D series: logogram 120
E series: *550, 552, 551, *89, *21f
Fa: HT 16, 20, 24, 38; Fb (HT 31
G series: logograms in heading: Ga VIN, Gb other logograms
H series: no logograms
X series: fragments

Packard's system is not easily applied to ZA & KH docs, which lack A, B, E, G, and F.
Schoep 2002 classifies the documents, especially those from Haghia Triada, into 5 main categories (I have included this information with the presentation of the individual documents):

Ia: Mixed Commodities, with heading and transaction sign (TS); these probably compile information from individual accounts
Ib: Miscellaneous Commodities, list(s) and numbers; these probably compile information from individual accounts
Ic: Miscellaneous Commodities, list(s) and numbers, and a word ("total," "deficit") and number at the end; these probably compile information from individual accounts
II: Specialized, one logogram (and variants)
III: Single Commodity, one logogram for the whole list
IV: No logograms, always whole numbers, never fractions; therefore probably a list of personnel (such lists may also appear in the previous types of documents)
Va: Combined commodities, a separate section of 100/102 VIR and word & number; again, always whole numbers, never fractions, and therefore mainly about personnel (and rations?: *303, figs, wine, OLE; e.g., HT 27, 89, 94, 100)
Vb: Combined word & number and list & number sections

It is likely that the mixed commodities documents compile information drawn from individual accounts, the lists of personnel relate to a census or work-force, and the single commodity documents summarize compiled information.

Many lists are preceded by a short heading of no more than 2 words (3 words on HT 96a, HT 117a), followed by a single sign (e.g., AB 04 TE, AB 28 I, A 307 =? AB 39) -- these are probably transaction indications (e.g., assessments, contributions, disbursements). There may be subheadings to sections of the document (HT 93a.3-4 and 4-5; HT 120.3-4; KH 7a.3; HT 27a.4-

5). PH 6 is unusual in that it presents 5 signgroups over 4 lines with NO ideograms or fractions. Longer, non-transaction, prose-like statements are rare; most of these constitute the presumed religious inscriptions, ZA Zb 3 (5 signgroups, 1 logogram), and the hairpins (ARKH Zf 9; CR(?) Zf 1; KN Zf 31; PL Zf 1).

In the lists, commodities are almost always listed by ideograms, single signs that stand for objects (occasionally the commodity is not mentioned, being implied somehow); logograms are often ligatured with syllabograms and/or fractions to denote aspects of the object.
Numbers then follow. Most fall into one of several categories: fractions, which may often indicate allocations (commodities leaving the administrative center in small quantities), the number 1 (probably personnel), whole large numbers (e.g., 20, 50, possibly alluding to assessments), and large random numbers (probably referring to contributions, either actual or in deficit).

6d. Balance Ledger Tablets

Balance Ledger tablets are transaction documents that record what appear to be contributions and disbursements (somewhat like "income" and "expenses"). They can be recognized by the fact that they present two lists of mostly the same NAMES. Original assessments (last column) can be deduced by totaling the contribution and the disbursement.

Here are two examples, HT 28 and HT 114.

HT 28
Since sides a & b carry most of the same names, it is likely that this tablet is in the form of a "Balance Ledger", with side a recording contributions and side b recording debits -- if so, then b.1: U-MI-NA-SI probably means "owes" (vel sim.; cf. HT 117a.1-2: MA-KA-RI-TE KI-RO U-MI-NA-SI. The words KI-RI-SI [TY 3b.1] & KI-RI-TA_2 [HT 114a.1] seem to be verbal variations on KI-RO, "debit" [vel sim.]).

name	a: contribution	b: U-MI-NA-SI	assessment
A-SI-JA-KA			
GRA+QE **5** JA-QI[f]	OLE+U OLE+KI **2** OLE+MI L_2 OLE+TU **1**		5 2 L_2 1
	VINa **6**		6
SA-RA_2	OLE+DI **1**	OLE+DI **5**	6
	NI **2**	NI 2	4
	VINa **3**	VINa **4**	7
		GRA **20**	20
	VIR+KA VINa **6**		6
A-RU-DA-RA	GRA **5**		5
	*304 **2**		2
	OLE+DI **3**		3

I-TA-JA	OLE+DI **10**		10
PU-RA$_2$		NI **6**	**6**
WI-DI-NA		OLE+DI **3**	**3**
		VINa **3** E	**3** E

Another arrangement, by commodity, reveals the ratios:

commodity	name	contribution	U-MI-NA-SI	total
VINa	VIR+KA SA-RA$_2$ WI-DI-*NA* JA-QI^f	**6** **3**	**4** **3** E **6**	**22** E
OLE+DI	SA-RA$_2$ A-RU-DA-RA I-TA-JA WI-DI-*NA*	**1** **3** **10**	**5** **3**	**22**
GRA	A-SI-JA-KA A-RU-DA-RA SA-RA$_2$	**5** **5**	 **20**	**30**
NI	SA-RA$_2$ PU-RA$_2$	**2**	**2** **6**	**10**
OLE+?	JA-QI^f	**3** J L$_2$		**3** J L$_2$
*304	A-RU-DA-RA	**2**		**2**

The ratios seem to be as follows:

VINa+OLE = 44E
GRA+NI+OLE?+*304 = 45 J L2
or: VINa+OLE =? GRA+whatever

HT 114 can also be rearranged as a Balance Ledger -- here, KI-RI-TA$_2$ is a likely variant on KI-RO (cf. KI-RI-SI on TY 3b.1), perhaps a verbal. If so, then side a lists what is owed, and side b lists what has been contributed (SA).

name	a: KI-RI-TA$_2$ (owed)	b: SA (paid?)	assessment
SA-RA$_2$	a: GRA **10**		10
	a: VINa **1**	b: SA (paid?) VINa **9**	10
	a: OLE **7** NI **1** BOS^m **3**		11 oil, figs, cattle

Again, the ratios appear to be similar in proportion to those in HT 28:

GRA = 10, VINa = 10, OLE+NI+BOS^{m} = 11
or: VINa:GRA:OLE+BOS^{m} = 1:1

7. Decipherment

My own aim in producing these files has NOT been to decipher Linear A.
Most scholars have tried one of two approaches to identify the phonetic values of the signs: the "acrophonic" principle, and using vocabulary to identify a language -- I don't believe either works very well.

<u>The acrophonic process</u>
Decipherments based on reading the signs as pictograms, then identifying what the object was called in a language, and then identifying the phonetic value of the sign as the initial sound or first phoneme of the object's name (the acrophonic principle) -- this process does not seem to work for Linear A for two major reasons.

1, the identifying term for the "pictogram" cannot be proved in advance of deciphering the script.
2, it can be demonstrated that, for several Hieroglyphic & Linear A signs, the acrophonic principle probably did not operate. Hieroglyphic *012 , a bull-head, becomes Linear AB 23 MU, and Hieroglyphic *018 , a dog head, becomes AB 60 RA. My guess is that the phonetic value of these signs reflect the sound the animal makes, "moo" and "arf" (in English). And there are other examples where the **sound** of the object seemingly relates to its phonetic value (e.g., Hiero *057 , a key sistrum, becomes AB 67 KI [the clinking sound of a metal rattle]).

This is not to say that the acrophonic principle is <u>never</u> appropriate to Linear A. Valério 2007 demonstrates that the word for master/lord is DU-PU$_2$-RE and that the first sign DU A51 is based in form on the Egyptian *sr* (Gardner A21), "official/dignitary/courtier."

<u>Using vocabulary to identify a language</u>

For me, vocabulary does not necessarily identify a language (English, for instance, has a large German, French and Classical Greek and Latin vocabulary); grammar would identify a specific language more securely. Thus, I am not immediately swayed by the process of identifying words in another language as Linear A words (e.g., KU-NI-SU in Linear A as the Semitic term for emmer wheat) -- this is not to say that I don't find such correspondences impressive and interesting. Compare Nakassis and Pluta 2003: 335: "A number of scholars have attempted to decipher Linear A, identifying it with known languages such as Semitic, Luwian, and even Greek. These studies begin by attempting to etymologize a small number of individual words, largely ignoring overall context."

My own method has been strictly internal, to examine the texts as accounting documents, and to use the numbers to identify transaction terms and patterns in vocabulary, and then to pay special attention to vocabulary variations, especially in prefixes and suffixes, in order to tease out a grammar.

Whatever language Linear A turns out to be (Semitic, Indo-Hittite, Greek, or Martian), will be fine with me; I have no set predisposition.

8. Language

Linear A has not yet been demonstrably linked to **any** known language family.

"The languages which have been used for comparison are of two families: Indo-European, especially an Anatolian language such as Luwian (Palmer, Meriggi [and Ed Brown of UNC-CH]); Semitic (Gordon, Best, and others)... First no inflexional forms such as characterize Indo-European or Semitic languages can be clearly demonstrated, hence the identifications depend largely on vocabulary, which is notoriously easily borrowed. Secondly, the Semitic comparisons are mainly with triconsonantal roots -- yet if the vowels are ignored we are leaving out half the information presented by the script, and thus much decreasing the chances of success. Thirdly, if the languge of Linear A does not belong to a well-known family, then the chances of identifiying it are virtually nil. This is not to say that Linear A remains undecipherable; as more documents are found and published, we shall understand more of it. But I doubt very much if speculation at this stage can help; I feel strongly that it is likely to belong to an unfamiliar type." (Chadwick 1975: 147)

If Crete was deliberately colonized in developed Neolithic, probably from SW Anatolia (Broodbank & Strasser 1991), it would seem logical to surmise that the Minoan language could be related to one of the Indo-Hittite dialects, most probably Luvian.

It has been recognized that Linear A contains a high number of affixes (prefixes & suffixes; Duhoux 1978), suggesting Linear A is "agglutinative rather than conjugating." There is a high number of prefixes (59% of the words Duhoux singled out; Linear B has 12%), playing an important role "in expressing gender, case or derivation" or some other kind of syntactic relation (Schoep 2002, 45-46).

Classical inscriptions from East Crete (Dreros, Praisos, Itanos) have been labeled Eteo-Cretan (Duhoux 1982); these may reflect an indigenous language.

Phonetic values of the signs (Godart 1984, amplifying Olivier's previous list)

certain: DA, I, JA, KI, PA, PI, RO, RI, SE, SU, TA, O
possible: TE, A, KO, RA

possible recent identifications

***34**

Sign *034 ⊂ has been suggested by several scholars to represent MNA (or, if a disyllabic value can be accepted, MINA), based on its resemblance to the crescent moon (Pope and Raison 1978, 28; Packard 1974, 107; Furumark 1956, 24).
Arguments in favor of this idea:

U-*034-SI (HT 15.1, 140.1, 2) =? U-MI-NA-SI (HT 28b.1-2, 117a.1-2)

PI-*034-TE (HT 116a.4) =?]PI-MI-NA-TE (AP Za 2.2)

Also see the note to *325, below.

Melena 2014:95, however, proposes a different phonetic value for the sign when it's used in Linear B. First, signs *34 and *35 (a reversed *34, now deleted due to its few occurrences) represent the same sign and, second, that *34 represents a possible aspirated /*a*/ or /*ai*/, which he designates a_5.

He compares *34-*ka-te-re* at PY Va 15 to *a-ke-te-re* PY Jn 832, a_2-*ke-te-re* KN V 118, and *ja-ke-te-re* PY Jn 11

***22=^{m}PI$_2$** (see Duhoux 1984, Janda 1986, Melena 1987; Tosa 2010)

***29=^{m}PU$_2$** (see Duhoux 1984, Janda 1986, Melena 1987; Tosa 2010)

***56=PA$_3$ or ^{m}PA$_3$** (see SMID 1981, p. 61; Duhoux 1984, Janda 1986, Melena 1987; Tosa 2010)

***65=JU** (see SMID 1981, p. 61)

***66=TA$_2$=TNA** (Pope-Raison 1978: 28).

Cf. KI-RE-*66 (HT 85b.1-2, HT 129.1) and KI-RE-TA-NA (HT 2.3, HT 108.1, HT 120.4-5); and *66-TI-TE (PK 1.3) and TA-NA-TI (HT 7a.4, 10b.4, 98a.2)

***304 = KA$_2$; ↑ *306 = A$_2$ (shape resembles AB 43, known from MY Zf 2); *318 = DI$_2$**

compare

HT Wc 3017	HT 94
*304+PA	a.1: KA-PA
DI-*306	a.4: *318-*306

QA-KU-RE	b.2: KE-KI-RU

*304 = KA	
JA-*304[(PH 14a)	cf. A-SI-JA-KA (HT 28a.1, 28b.1-2)
]*304+PA-DA-*047-KU[	
*304+PA (lots)	KA-PA (HT 6a.1; HT 8b.4; HT 94a.1; HT 102.1; HT 140.5)
*304+PA+*316+D3 (HT Wa <1021bis>)	
*304+PA-KU-PA (HT We 1020a)	

*306 = A	
]*306-JA-PI (ARKH 3b.1)	WA-JA-PI-[ ] (HT 9b.1)
]*306-KI-TA2 (HT 122b.2)	A-*301-KI-TA-A (TY Zb 4)
]*306-QE-DU[(KH 21.3)	
]-*306-TI-KA-A-RE[(HT 4.1)	A-TI-KA (ZA Wc.a1-2)
*306-TU-JA (HT 115b.3)	cf. JA-TO-JA[(ZA 4a.2-3)

***314 = PU_3**

Valério independently; Owens 1999 & Facchetti 1999a, 132 identify the value as BU, a variant of PU_2; so as not to start a new consonant "row," I conform the sign to PU_3

***325 = MA-NA?**

If prefix I-/J- indicates a dative, causing A-SA-SA-RA to change to JA-SA-SA-RA-ME (IO Za 6, 12, 16; PL Zf 1; PS Za 2; TL Za 1b) or to JA-SA-SA-RA-MA-NA (KN Za 10), then we might interpret RI-QE-TI-A-SA-SA-RA-*325 (PO Zc 1) as RI-QE-T plus I-A-SA-SA-RA-*325, identifying *325 as either ME or MANA, probably the latter, if U-*325-ZA (HT 10a.2, 3; HT 85a.3) stands for *U-MI-NA-ZA (a variant on U-MI-NA-SI?; see note to *34, above). Also cf. A-*325-ZA.

***348 (hapax legomenon) = SI_2**

Compare *348 *303KL_2 with SI *303 KL_2 (KH 11.3-4, 6 respectively)

***363 (and** ***364?) = SO_2**, so Valério; I agree (the sign derives from Hiero *043, from which B12 *so* derives); see Hiero #039 and parallel Linear A tablet HT 9.

9. Vocabulary

10a. Transaction Signs

Single signs indicate types of transactions; they are preceded and followed by dots (except on HT 44b.1, 63.1, 133.1). In Linear B, transaction signs can mean different things in different contexts (e.g., DA in PY En 609 or in Aa, Ab sets). Transaction Signs "provided information about the kind (or purpose) of transaction and that the format of a tablet indicated the use to

which a commodity was put by the administration (its administrative status), its provenance or its destination. Thus, the lists of mixed commodities are clearly drawn up to a different administrative purpose than the single commodity tablets: the former record a single origin or destination for many commodities whereas the latter record several origins or destinations for single commodities" (Schoep 2002, 141). The low frequency of most transaction signs implies they are context-bound, which would be appropriate for contributions; those occurring with KI-RO (e.g., HT 123) probably are also dealing with "incoming commodities or assessments of incoming commodities" (Schoep 2002, 142).

DI, a single sign with people (cf. HT 85b); on HT 20 it occurs with the fraction J (Schoep 2002, 139); since two of the words on HT 20 end in -JU, if that were like Linear B, meaning "son of," perhaps the fraction would appropriate for a child's ration.

KA, apparently associated with people or a group of people (title, collective, occupation); cf. HT 11b.2, 45a.2 (Schoep 2002, 137). The following discussion explores the possibility that VIR+KA are porters (see the individual notes to HT 28 and HT 88).

> On HT 28a.4 VIR+KA VINa 6 comes just after VINa 3; the note there suggests that VIR 6 might be responsible for the VINa 3, perhaps 2 VIR per half-unit of VINa or 2 VIR per pithos of VINa (they could slip two poles through handles and carry each pithos like a litter). On HT 88 VIR+KA 20 or 26 are listed agains FIC KI-KI-NA 7; perhaps the figs are rations for the VIR+KA, if 7 units of figs, that would be 1:~3. On HT 140 KA totalling at least 8 and possibly as many as 14 are associated with an amount of OLE+U that totals at least 6J, again perhaps rations (OLE+U : KA = ~2:1).
>
> HT 11b.2-6 lists a total of 180 *86 KA in 5 groups (**40, 30, 50, 30, 30**). Nodules HT Wa 1322-1466 are each inscribed with KA and impressed by 14 seals and nodules 1467-1470 are incised with KA but are unimpressed. The total of nodules with KA is 149. If KA is an abbreviation for VIR+KA, it is possible that the nodules are itemizing ca. 150 personnel individually that are listed in (the first 4 of the 5) groups on HT 11. (*Thanks to Julio Fontan-Tejeiro, February 2014)*

It is also possible to see all of the 180 VIR+KA on HT 11 apportioned on other HT tablets: HT 28a.4 VIR+KA 6, HT 88.1 VIR+KA 20 plus 6 (by the Continuity Principle, see section 18, below), HT 97a.1 VIR+KA 82, and HT 100.1]VIR+KA 58, totalling 195. Indeed, the VIR+KA 58 on HT 100.1 might include the 57 that are listed individually on nodules Wa 1323-1379, all of which are impressed by CMS II 6 no. 11, the seal most frequently used on nodules.

KI usually occurs with people (HT 118, pigs).

- KI-RA: HT 49a.8
- KI-RO? (Raison & Pope in Duhoux 1978: 47-48); cf. HT 118; and mentioned on HT 49a.7&8

KU can either stand for KU-RO (e.g., HT 40.4) or designate a group of people (title, collective, occupation) (Schoep 2002, 136-7).

PA should be an abbreviation for a term modifying people; it may be in opposition to *188 -- HT 8 has, on the recto, PA, verso *188; PA fraction may refer to *618 (*302+KI). HT 10.3 has two separate entries, one for PA and another for *301 (Schoep 2002, 135, 137, 138).

PA_3 appears on HT 9b, listing payments (Killen 1969 equates this with Linear B *a-pu-do-si*, received. HT 9a has TE, probably expected assessments or contributions) (Schoep 2002, 138-9). The same meaning can be deduced for PA_3 on HT 34.6.

RE refers to people (HT 27, VIR in the heading; HT 32.4); it precedes the logogram *305 (Schoep 2002, 137)

SA (HT 114b.1) or **SI** (HT 30.1) = paid?

TA may designate a group of people (title, collective, occupation) (Schoep 2002, 136-7).

TE/TI, "from/of" (Valério 2007), associated with agricultural products and people, usually in large quantities. So, assessments (for future incoming commodities) or inventories of commodities arrived (Schoep 2002, 100, 168 "more likely to represent a kind of contribution to the administration (e.g., taxes, levies, payments of loans, tributes etc.) rather than distributions"). This meaning seems supported at HT, with large quantities of wine (collections, rather than allocations) associated with TE; the total amount of TE GRA, however, is small, 1/10th the amount registered with SA-RA_2. This is common, appearing on 21 HT texts in headings or sub-headings [HT 67, 96] and relating to agricultural commodities: AB 30 (FIC, occasionally), AB 120 (GRA), AB 122, AB 131 (VIN), A 302 (OLE). TE and SA-RA_2 are mutually exclusive (Schoep 2002, 98). See further examples below.

TU =? redistribution: HT 49a.8

WI =? "not included" vel sim; HT 102.4

***86** (for the form , cf. Linear B PY Mn 111.1,7). On HT 140, 86 is probably a single sign, applying to the following sign-group associated with VIR, always with whole numbers (cf. HT 27a.2, 94a.1, b.5). It may designate a group of people (title, collective, occupation); cf. HT 11b.2, 45a.2 (Schoep 2002, 136-7).

***188** is positioned like a transaction sign but it can also occur as a single sign. It only appears in initial or final position of two-sign sign groups, so it probably is not a syllabogram; it may be in opposition to PA (see HT 8 recto PA, verso *188). *188+KU may be a logogram on HT 16.3, 20.5 (Schoep 2002, 135-6).

***301** usually refers to people, but on HT 110b.5 it is accompanied by a fraction and may refer to the figs above (Schoep 2002, 138).

***305** may designate a group of people (title, collective, occupation); cf. HT 11b.2, 45a.2 (Schoep 2002, 136-7).

***307** is positioned like a transaction sign but it can also occur as a single sign. It is associated with people, also when doubled (*638). From its appearance on HT 27 and HT 89, it can refer to allocations (Schoep 2002, 142).

***318** is associated with people

***326** is associated with agricultural commodities, e.g., figs (HT 91, this is the only mixed commodities tablet this sign appears on)

***346** is associated with people

***406VAS** is associated with people

***516, I+[?]** -- used with figs (HT 96, ZA 1); it may also designate a group of people (title, collective, occupation (Schoep 2002, 136-7). It might be related to prefix I-, meaning "to", implying an allocation (see section 13c, below); see, for example HT 96, where one section is headed TE and the next is headed *516 (also see DA-I under Transaction Words).

***523, TI+A** may designate a group of people (title, collective, occupation) (Schoep 2002, 136-7).

***545, KI+MU** may designate a group of people (title, collective, occupation) (Schoep 2002, 136-7).

***638** --- see *307

***656 (*341+PI)** is associated with people

9b. Transaction Words

A-DU = "assessment"?: HT 95.b1 (and elsewhere)

Heading **A-DU** (probably derived from Hieroglyphic *A-DE*).

From the examples below, Linear A's A-DU usually functions alone and as a primary heading to a tablet; it may be modified by transaction signs (e.g., TE) and/or logograms (e.g., VIR, GRA, OLE).

As a prefix, it usually is followed by -RE-ZA (which may be a word by itself [HT 88.1-2]); here again it is usually a primary heading, but in one abbreviated form it is simply an item in a list (PK 1.2). The abbreviation DU-RE-ZA occurs at least once (and supplemented with an additional suffix -SE), and as an item in a list. A-DU also occurs as prefix to another word, KU-MI-NA, which exists by itself (KU-MI-NA-QE [HT 54a.2 & HT Wc 3014a-b]) as well as on the same document as A-DU-KU-MI-NA, again as another item in the list, prefixed simply by A- two lines above (ZA 10a.1-2).

As part of longer words, A-DU may be prefixed by WI-N, possibly as a variation on the common prefix I-NA- (e.g., I-NA-TA-I-*79-DI-SI-KA [IO Za 5], I-NA-I-DA-[[IO Za 11], I-NA-JA-PA-QA [PK Za 11d], I-NA-WA [PH 6.1]).

A-DU as primary heading

A-DU	HT 88.1; HT 86a.4; HT 95b.1; HT 99a.1; KH 23.1
A-DU *638 VIR	HT 85a.1
A-DU • TE GRA+DA	HT 133.1-2
TE • A-DU • GRA	HT 92.1-2
A-DA-KI-SI-KA • A-RA-U-DA	KH 5.1-2
A-DU[	KH 23.1

A-DU as heading

A-TI-KA A-DU KO	ZA Wc 2
A-DA OLE+U	TY 3a.5

A-DU-RE-ZA

A-DU-RE[	primary heading: KH 4.1
A-DU-[•]-ZA *303	primary heading: KH 11.1
A-DU-ZA	item in list: PK 1.2
JA-DU-RA-TI	primary heading: KN 1b.1-2
A-DU-[••] OLE+KI	heading: TY 3a.3
DU-RE-ZA-SE	item in list (ZA 10a.5 & b.1-2)
]DU-RE-ZA-SE-[	?heading: ZA *20.1-2
]DU-RE-ZA	item in list (KH 20.4)
RE-ZA	item in list (HT 88.1-2)
]ZA-*321 OLE+KI	primary heading: TY 3a.1

A-DU as part of other words

A-DU-KU-MI-NA	item in list (ZA 10a.3-4)
DA-DU-MA-TA • GRA	primary heading: HT 95
WI-NA-DU	item in list: KH 5.3
]KO-A-DU-WA OLE+[	item in list: TY 3a.6

Two documents, HT 95 & HT 86, give further clues for understanding the meaning of A-DU; each records two lists, probably of place names. In the heading on the recto, DA-DU-MA-TA • GRA (HT 95a.1) is juxtaposed to the secondary heading on the verso, A-DU (HT 95b.1); HT 86 juxtaposes the heading A-KA-RU on the recto with A-DU on the verso (cf. KA-RU [HT 97a.1]). See the discussion of these two documents below: the words A-DU, A-KA-RU, and DA-DU-MA-TA, KA-RU seem to be transaction terms; I suggest something like "assessable", "payable" and "assessments", "payments". Uchitel, A., and M. Finkelberg, "Some Possible Identifications in the Headings of the Linear A Archives," *SMEA* 36 (1995) 29-36, suggest "let them do" for A-DU.

	HT 95 (GORILA I: 154-155)				HT 86 (GORILA I: 134-35)	
a.1	DA-DU-MA-TA • GRA			a.1	A-KA-RU	
a.2	DA-ME	**10**				
a.2	MI-<u>NU</u>-TE	**10**				
a.3	SA-RU	**20**		a.2	SA-RU	**20**

a.3-4	KU-NI-SU	**10**		a.1-2	KU-NI-SU GRA+K+L2	**20**
a.4	DI-DE-RU	**10**		a.3	DI-DE-RU	**20**
a.4-5	QE-RA2-U	**7**		a.3	QA-RA2-WA	**10**
b.1	A-DU			a.4	A-DU	
b.1	SA-RU	**10**				
b.2	[•]					
b.2	DA-ME	**10**		a.4	DA-ME GRA+B	**20**
b.2-3	MI-NU-TE	**10**		a.5	MI-NU-TE	**20**
b.3-4	KU-NI-SU	**10**				
b.4	DI-DE-RU	**10**				
b.4-5	QE-RA2-U	**10**				

HT 86 assigns 20 commodities to names (places?) in two lists headed by A-KA-RU and by A-DU.

On HT 95 these same names occur in each of two lists, headed by DA-DU-MA-TA and again by A-DU; in each list, however, most of the names have 10 commodities -- except: SA-RU has 20 in the DA-DU-MA-TA list and an additional 10 in the A-DU list, and QA-RA2-WA, for which 10 commodities are recorded on HT 86, is recorded (and spelled apparently the more common way, QE-RA2-U [cf. HT 1.1-2-]), with 7 in the DA-DU-MA-TA list and 10 in the A-DU list. For this odd number 7, cf. QE-RA2-U 197 (HT 1.1-2).

By allocating the names to separate lists, HT 86 looks like it is dividing the names into some kind of mutually distinct groups (e.g., by geography); HT 95, however, by recording all names in both lists, appears to be recording transactions (e.g., payment and non-payment or income and reallocation).

If DA-DU-MA-TA and A-DU are related (e.g., "contributions" and "not paid"), then compare A-KA-RU with A-KA-RU 20 (HT 2.1, the total of the next two numbers), and with KA-RO[(HT 71.1) and KA-RO (HT 97.1) -- on this last document, KA-RO 82 may record a total of the numbers recorded against place names: *327 33, KA-NU-TI 25, PA-I-TO 6, DI 4, NA-TI 4, MA-DI 5, TA-TI 2, DE-[•] 3.

DA-DU-MA-TA GRA = "grain contributions"?, if DA-DU-MA-TA is related to A-DU q.v.: HT 95.a1.

DA-I = "total"?: HT 12.6 (Schoep 2002, 162); cf. DA-I-PI-TA, ZA 8.5. If DA- (as in DA-DU-MA-TA) indicates in some way a completed action (like a perfect of A-DU), could DA-I be a completed transaction *516 I+[?] (q.v. under Ideograms/Logograms)?

E-*82 = "assessment" or "paid"; ZA 4

KA-I-RO = "balance": ZA 8.6

KA-PA occurs only in the HT tablets; it seems to be a transaction term or heading on HT 6 (FIC coming TE "from" a list of names), HT 102 heading a list of GRA from five names, HT 140.5 where the word seems to conclude the list. Ruth Palmer 2004 suggested that *ka-pa* in Linear B

meant processed "fruits" (cf. LinB *ka-po*). But KA-PA also seems to be a name (personal or place) on HT 6a.4-5 ("KA-PA-QE" in order to distinguish it from the heading), HT 8b.4 listing PA in small amounts (therefore "to"?) and a list of names, HT 94a.1 where contingents of KA-PA VIR are distinguished from rations listed against SA-RA_2, HT 105.1 where VIR is being contributed from KA-PA and SA-RA_2 in almost equal numbers. Schoep 2002, 166, thinks that KA-PA might mean "summary account" vel sim., or a place name.

KI-RA = "balance"?, a transaction term on HT 103.5 (Schoep 1994-5, 71, n. 60); cf. ZA 8.1

KI-RO = "owed", "deficit" (Younger 2003)

- "balance": HT 1
- "itemized payments/debts": HT 88.4, 93b.1, 94b.1
- "owed" (Hooker 1975; Duhoux 1989, 79): HT 30.4, 123a; HT 118; cf. HT 49: a.1-7 totals 10; if KI<-RO> 1, then 9, which is what a.8 records (5+4), with the KI<-RO> 1 repeated
- KI-RO is also mentioned: HT 88.4, 124
- verbal forms of KI-RO; also see below, "Transaction Words"
  - KI-RI-SI -- 3rd singular (TY 3b.1)
  - KI-RI-TA_2 -- 3rd plural (HT 114a.1)

KU-PA, possibly a transaction term on ZA 11a.5 (Schoep 1994-5, 67, n. 47); cf. HT 110.2

KU-RA looks like a totaling word on ZA 20.4

KU-RO = "total" (Schoep 2002, 160-62; Younger 2003)

- secure: HT 9.a & b, 11.b, 13, 25.b2-4, 85.a, 88.4-6, 89.4, 94.a3 & b1-4, 104, 117.a1-6, HT 118 (with 5 having been omitted), 122?, 123.a, 127.b4-7; ZA 1?, 15, 17; cf. HT 116.
- with restorations: HT 27a.1-7, 100, 102
- rounded off: HT 119
- also mentioned: HT 39.5, 40.3, 46a.2

Perhaps the word is related to Semitic kl. "whole". (see Hooker 1975) or the Greek 'kolon', with the often Linear A 'u' substituting for Linear B 'o', and thus meaning 'sum'). Since KU-RO does not appear in Linear B, it could be a Minoan word (but see comment on PO-TO-KU-RO)

PO-TO-KU-RO = "grand total" (Palmer 1995; Schoep 2002, 163): HT 122b.6 (grand total of personnel; off by one?), 131.4 (mixed commodities, so it must total some common denominator, personnel?). If KU-RO is Minoan, is PO-TO- also, or is it derived from IE, meaning "power" (vel sim.), thus PO-TO-KU-RO = "power total"?

U-MI-NA-SI = "owed"?: HT 28b.1

10c. Place Names

A few place names can be identified (with more or less [??] plausibility) with actual places:

DA-U-*49 =? Linear B *da-wo*, Ayia Triada (Palaima 1994)

DI-KI-TE (several occurrences) - Dikte (Mt Dikte in east Crete, from the numerous mentions of DI-KI-TE in the PK documents; less likely to be Mt Ioukhtas [Owens 1993b])

I-DA (several occurrences) - Mt Ida

I-TI-NI-SA (ZA 15a.3) - Itanos (north of Palaikastro)

KU-NI-SU = likely a place-name (not "grain"; Hooker 1975; Duhoux 1989, 79; Schoep 2002, 148-50); "Knossos" is very doubtful.

The word appears only 3 times, and probably within the region of south Crete: as a heading without numbers (therefore a place?) on HT 10.1 (therefore nearby?); as a subheading on HT 86.2 & HT 95.a.3-4 [assessments?].

KU-TA[(HT 115b) might be Linear B *ku-ta-to*

KU-*79-NI (HT 13, HT 85a), perhaps Linear B *ku-do-ni-ja* (Bennet 1990, 193-211); KA-U-*79-NI cannot be a spelling variant (Schoep 2002, 156)

PA-I-TO (HT 97, HT 120) - Phaistos

SE-TO-I-JA (PR Za 1.b) - Archanes? (Owens 1994b), Ioukhtas?

SU-KI-RI-TA (PH Wa 32) - Sybrita (or Sygrita, in the Amari valley, modern Thronos)

TU-RI-SA (KO Za 1b-c) - Tylissos

SA-RA$_2$ (only in the HT tablets) - Since SA-RA$_2$ never appears on a document with transaction term TE, Schoep (2002, 98, 165-168) thinks it might be a transaction term. Other than that, however, SA-RA$_2$ seems to operate like any other NAME, presumably a toponymn since it occurs so frequently. Some 12 tablets (HT 18, 28a+b, 30, 90, 94, 99, 100, 101, 114, 121, 125, 130) associate SA-RA$_2$ with a set of 4 commodities (*303, FIC, VINa, GRA, OLE), usually in that order when they occur together. But SA-RA$_2$ is occasionally listed against other commodities: BOSm (HT 114a.3-4), VIR+KA HT 28a.4); HT 30, 32, 33, 34 seem to form a separate set of SA-RA$_2$ documents, listing QA+[?]+PU, QA2+[?]+RE, MI+JA+RU, MI+JA+KA, E+KA, *305, *341+PI, PA$_3$+QE. The quantities are often modest (although HT 105 lists SA-RA$_2$ VIR 125, and HT 102 lists SA-RA$_2$ GRA 976), perhaps reflecting disbursements to SA-RA$_2$ (perhaps that is implied on HT 93 which lists ASE I GRA+PA 26, then SA-RA$_2$ 20, then QA-QA-RU I GRA+PA 5, as if disbursements to ASE and QA-QA-RU must be spelled out ("I") but they are automatically presumed with SA-RA$_2$.

9d. Other Words

TA-JA =? "five" (Olivier 1992)
DU-PU$_2$-RE = lord/master (of a place, e.g., DI-KI-TE-TE-DU-PU$_2$-RE, Lord of Dikte; Valério 2007)

KI-KI-NA (HT 88.2) = "figs" (fresh or dried, green or black)

PU-KO = "bronze"?; see the commentary to HT 31

10. Ideograms/Logograms

The attach folder ("Linear A Ideograms") gives the common, identified ideograms and their signs, and copies of GORILA's sign charts and palaeographic sign charts

The administrative documents with logograms seem chiefly to concern people (A101/102 VIR people = B 100; A 352 might be the equivalent of B *102 MUL woman), agricultural products (AB 30 NI figs, AB 120 GRA grain [probably barley, Palmer 1989, 1992], AB 122 OLIV olives, AB 131 VIN wine, A 302 = B *130 OLE olive oil), and less frequently livestock (AB 21 OVIS sheep, AB 22 CAP goats, AB 23 BOS cattle, AB 85 SUS pigs), cloth (A 54 [occurring as a logogram 5 times] =? B *159) and vessels (A 400VAS-A 418VAS). Other ideograms are rare (A 308-A 371), most occurring only 1 or 2 times. Linear A does not have logograms for arms and armor (except AB 191 GAL helmet), spices (AB 123 = *123 AROM is only a phoneme in A [Hiero *157 occurs on seals #291 and CMS II 3.23 as a logogram]), or metals (unless A 327 [; HT 97a, HT 119] = B *140 AES bronze/copper).

***01 (DA)**, logogram on HT Wa 1031
Ligature

- DA+RE+SE (*502)
- DA+RO (*501)
- DA+*301 (*503)

***02 (RO)**, on HT Wa 1032-1121, Wb 2001, 2002
Ligature

- RO+RO[(*504)

***03 (PA)**, common

***04 (TE)**, common
Ligature

- TE+RO[(*505)

***07 (DI)**, common
Ligature

- DI+QE (*506)

***08 (A)**, common

***09 (SE)**, HT Wc 3004b, 3005b

***13 (ME)**, not a logogram by itself
Ligature

- ME+VINa (*507), honey wine?

***16 (QA$_2$)**, not a logogram by itself
Ligatures, *508-*511 occur with agricultural products (with *303, wine, oil, etc.) in whole numbers and fractions

- QA$_2$+[?]+PU (*510)
- QA$_2$+[?]+PU+RE (*511)
- QA$_2$+[?]+RE (*508)
- QA$_2$+[?]+RE+PU (*509)

***17 ZA**, HT 123b.1, ZA 1a.2

***20 ZO**, KH 57.2

***21 (QI)/OVIS** = probably sheep (Palmer 1995); also as a syllabogram (OLE+OVISf on TY 3). HT 38: OVIS no sex indicated = castrated males (wethers), producing finer wool and in greater quantities (Halstead 1990-1, 343-65). Schoep 2002, 122, cites low numbers of sheep (ZA 22: 100, otherwise no more than 30), but we can add now 53 sheep cited on the Thera tablets. Still, this is not much evidence for large scale textile production.
Ligature

- QI+SI (*512) or OVIS+SI, cf. Linear B SUS SI (for *si-a$_2$-ro*, fattened or to be fattened)

***22/CAP (B *107)** = probably goats (Palmer 1995); also as a syllabogram (Duhoux 1984: 61 n. 38 suggests a phonetic value of MI?). HT 20.3: CAPm F, probably an animal product because of the fraction.

Ligature

- CAPm+KU (*513)

***23 (MU)/BOS** = cattle. *23 appears by itself (HT, ZA, KH 5.6) and may be "ox" or possibly "cow," although the sign lacks the split stem that Linear B 109b has. The masculine variant (*23m, with stem crossed twice) appears frequently (GO Wc 1b; HT 30.4, 5, 114a.3, 121.3; KH 87.3, Wc 2069) (Palmer 1995)

***24 (NE)**, HT Zd 156?

Ligature

- NE[ ]KI (*514)

***26 (RU)**, HT 12a.2, b.2

***27 (RE)**, PH 15a; ARKH 4a.2; HT 27a.2, b.2, 5, 6, 32.4, 41a.3

Ligature

- RE+SE (*515)

***28 (I)**, HT 43.2, 115a.1, b.3, Wa 1150-1174; *28b: HT 62[+]73.3; ZA 4a.5-6, 5b.1, 15a.4-5, 6b.1

Ligature

- I+[?] (*516), "to"?
- I-GRA+PA (*517)
- I-OLIV (*518)
- I-*301 (*519)
- I+*301 (*520)

***30 (NI)/FIC** = figs (common) (Palmer 1995); one of "the Aegean triad" (along with *120 GRA, *122 OLIV; Schoep 2002, 98). As in Linear B, olives and figs hardly ever appear

together (exc. HT 44, 91, 131). Incoming quantities are probably from palatially-owned orchards (HT 123 lists other commodities, so presumably these olives are coming from other sources); out-going quantities are usually small, even fractional (Schoep 2002, 96).

HT 88.2 presents the only certain occasion when the word for a logogram is spelled out (the word appearing after the logogram; another possible occurrence would be *188 [Talent?] MI-NA on KN Za 19): FIC • KI-KI-NA **7**. This is odd, since NI may be the acrophonic symbol for "nikuleon," an old word (Minoan?) attested by Hesychius for "figs" (Neumann 1962). Perhaps KI-KI-NA distinguishes fresh from dried or green from black figs, meaning one or the other, depending upon which would be otherwise automatically assumed.

***31**, SA, perhaps a logogram for *SA-SA-ME?; cf. HT 23a.4-5 SA-SA-ME, and PH 16b.1, HT 97b (cf. Linear B *sa-sa-ma*)

Ligature

- SA+MU+KU (*521) or SA+BOS+KU
- SA-VINa (*522), sesame wine?

***37 (TI)**, common

Ligature

- TI+A (*523) or A+TI?
- TI+*412VAS (*524)

***38**, E, both a syllabogram and a common logogram (at HT, KN). E by itself does not occur with *526, and they occur in different amounts, E in whole numbers. JGY: as a logogram, E often appears with whole numbers (e.g., 3, 5) amongst a list of commodities in fractional amounts (e.g., HT 60, 91). KN 1 lists E 240 and E 105.

Ligature

- E+[ ] (*525)
- E+KA (*526), in fractions (Schoep 2002, 133)

***39 (PI)**, logogram only in ligature

Ligature

- *341+PI (*646)

***40 (WI)**, HT 102.4

Ligature

- WI+ZE[(*527)

***41 (SI)**, common
Ligature

- SI-DI (*528)
- SI+ME (*530)
- SI+ME-KI (*531)
- SI+SE (*529)
- SI+TA_2 (*532)
- SI-CYP (*533)

***44 (KE)**, only in ligature
Ligature

- *406VAS+KE (*656)

***47 (?)**, MA 1a,b

***50 (PU)**, HT 34.7

***51 (DU)**, ZA 10b.7
Ligature

- DU+[ ] (*534)

***54 (WA)/TELA** = cloth, HT 16.2, 20.4, Wc 3019. Without ligatures, TELA is measured in fractions (HT 16, 20, Wc 3019?), perhaps by weight (cf. HT 16, 20: *188+*81).
Ligatures

- TELA+KU (*535), counted in units, associated with *188
- TELA+TE, Tel Haror only (cf. Linear B TELA+TE = *tepa* cloth, a Minoan word; Oren 1996, 105)
- TELA+SE, Thera only
- TELA+*312 (*536), also TELA+KU?, counted in units, associated with *188 (Bennett 1975: 61, & Melena 1975: 108-10 both give TELA+ZO?; Younger 2005 identifies A312 as KU [cf. Hieroglyphic document]

***56 (PA_3)**, HT 9b.1, 132.2, 34.6

Ligature

- PA$_3$+QE (*537)

***57 (JA)**, PH 14a, 16b.1; HT Zd 156?

Ligature

- JA+KA (*529)
- JA+RU (*538)

***58 (SU)**, KH 11.2, 22.1

Ligature

- SU+MI (*540)

***59 (TA)**, MA 4a; ZA 10b.6; HT 94a.2, Zd 156(?), WA 1267-78

***60 (RA)**, logogram only in ligature

Ligature

- RA+KA (*541)

***61, O**, HT 113.3, Wa 1279-81; KH common

***65, JU**, HT 97a.4, ZA 11a.4; Linear B FAR?, perhaps a logogram, although it may occur as such only once, on ZA 11a.4 followed by a fraction

Ligature

- JU+*317+QE (*542)

***66 (TA$_2$)**, HT 11a.5

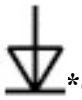
Ligature

- TA$_2$+CYP (*543)

***67 (KI)** = KI-RO, common

Ligature

- KI+MA+RU (*546)= KI+LANA
- KI+MU (*545) or KI+BOS
- KI+[ ] (*544)



***69 (TU)**, HT 49a.8; KH 91.3; ZA 12b.1

Ligature

-]TU+RO (*547), cf. Linear B $TURO_2$, "cheese"

***73 (MI)**, logogram only in ligature

Ligatures (*550 and *552 are found in different contexts than *551; *550 is measured in fractions, while *551 is measured in whole units [Schoep 2002, 130])

-]MI+JA (*548 , MI+JA on PH 3b.1 (3a concerns wool)
- MI+JA+KA (*552)
- MI+JA+I (*551)
- MI+JA+RU (*550)
- MI+JA+[ ] (*549)
- MI+*301 (*553)

***74 (ZE)**, HT 16.4, Wa 1282-132; KH Wa 1005-1010

***77 (KA)**, HT 85b.3, 140.4, 11b, Wa 1322-1470, Wc <3018>a

Ligature

- KA+A (*554) or A+KA?

***78 (QE)**, logogram only in ligature

Ligature

- QE+GRA+PA (*555)

***79 (WO_2??)**, HT 66.1
Melena 2014, 83, 89 suggests *wo_2* for Linear B *79.

***80 MA**, perhaps a logogram on HT 146.3, 110b.2, 5 (cf. Linear B *ma-ra-tu-wo*)
Ligature

- MA-RU (*80-*26,), wool
-]MA-RU (*558)
- MA+RU (*559)
- MA+RU+ME (*561)
- MA+RU+RU (*562)

Linear A thus has no separate logogram for LANA (unlike in Hieroglyphic *84 and Linear B *145). The ligatures appear on a few documents (HT 12.4-5, HT 24a.1-5, KH 43.1, PH 3a.3) and the word is actually spelled out, MA-RU on HT 117a.3 and]MA-RU-A on TY Zg 1; MA-RU does not seem to appear in Hieroglyphic. Linear B LANA *145 pictorializes the Linear A ligature, perhaps being influenced by the pictorial Hieroglyphic sign *084 . The classical Greek word "μαλλός, mallós," wool, may thus be a loan-word from Linear A (and Hieroglyphic?). At PH, LANA is measured in small quantities, possibly disbursements; at HT, the quantities are large, possibly collections (Schoep 2002, 132).

***81 KU**, common (cf. Linear B *ku-mi-no*, a Semitic word, Chadwick 1975, 119). It occurs in whole numbers on HT 127b.4, but in fractions on HT 45b.4, 5
Ligature

- KU+[ ] (*564)

***85 AU/SUS**, HT 38.2, 118.1; PH(?) 31a.3, b.4; pigs (Palmer 1995);
Ligature

- 85+SI+RE, SUS+SI+RE or SI+AU+RE, the Linear A predecessor of Linear B *si-a_2-ro*, fattened?

***86 (?)**, HT 11b.2, 27a.2, 94a.1, b.5
Ligature

- *86+'*188' (*565)
- *86+*188 (*566)

***87 TWE**, only on HT 126b.2, 3

***100/*102/VIR**, common = person (Palmer 1995); also a syllabogram (SI-VIR-[•] on HT 72.1, heading; VIR-I on HT 11a.4 & HT 93a.5-6 as a name in a list; VIR-*329 on HT 128b.1 as a name in a list).

There does not seem to be a common logogram for woman (MUL), but cf. VIR+KA (below) and *352 (KH 99; KH Wc 2100).

Ligatured (connoting aspects, like occupation, age, gender, status? Schoep 2002, 113)

- VIR+KA (*568), women? Godart 1984, 125 (cf. HT 28)
- VIR+*307 (*569)
- VIR+*313a (*570)
- VIR+*313b (*571)
- VIR+*313c (*572)
- VIR+[?] (*567)

On HT 85a, VIR are sorted into sets of 6. HT 105.2 has VIR 234 or 39 sets. Compare HT 88.1-2 RE-ZA VIR 6 and .4-6 6; HT 89.3: VIR+313 13, TA-RA 5 = 18 (3 sets); HT 93a.5-6: VIR-I 6; HT 94a.2 VIR+313b 18; HT 97 *638 VIR+KA, the numbers total 192 (32 sets) - also PA-I-TO contributes 1 set; HT 127, VIR+313 24 (4 sets); HT 128b.1 VIR-329 GRA+KU 6; KH 7a.4-5: KU-*-KO-E VIR+313b 18 (3 sets); KH 26.a VIR 60 (10 sets); KH Wc 2004, VIR 6; KH Wc 2029, VIR+KA 6; KH Wc 2030, VIR+KA 6; KH Wc 2106, VIR KA 6; PE 1.3-4, VIR 72 (12 sets); TY 2.3-4, VIR 84 (14 sets).

Contrast HT 94 a.1: VIR 62, 94b.5 KU-RO 5; HT 100, whose total VIR is 97 (1 more than 16 sets); HT 102, VIR GRA+PA 33 (11 half-sets?); HT 105.3, SA-RA$_2$ VIR 235 (one more than 39 sets); HT 119: VIR 68 (not associated with a name), 11 one-third sets; KH 7a.2-3, 4, VIR+313, 10 & 4 respectively; KH 25, VIR 140 & VIR 10; KH Wc 2031, VIR+KA 4; KH Wc 2032, VIR+KA 9; KH Wc 2100, VIR-352-JA 9; PE 1.2, VIR 53 (restored, one less than 9 sets); PH 8, VIR 11 (one less than 2 sets).

Analysis of HT 85 (thanks to Brent Davis for relating side b to side a):

- On HT 85a, the numbers total (KU-RO) **66**, or 11 sets of **6** VIR each. The 11 sets are specified on side b, being listed against one NAME (and one or two logograms associated with a name). These groups of 6 personnel are obviously conventional and may reflect small, easily supervised sets of workers.
- The names on side a recur frequently in the HT corpus, notably HT 10 (DA-RI-DA, U-*325-ZA [twice], DA-RE [twice]); HT 122 (DA-RI-DA, DA-RE, PA^3-NI; cf. PA^3-NI-NA on HT 6 and HT 93), and HT 13 (DA-SI-*118, KU-ZU-NI). This frequency probably implies that the names on HT 85a are toponymns.
- In contrast, the names on side b recur less frequently: WA-DU-NI-MI on HT 6, RE-DI-SE perhaps as RA-TI-SE on HT 6, KR-RE-TA^2 on HT 129, and the disyllabic names ME-ZA on HT 10, TE-TU on HT 7 and HT 13, QA-*310-I on HT 8 and HT 122, and MA-DI on HT 3, HT 97, HT 118, and PH(?) 31a. These names may thus be personal names, that is, the names of individuals.
- If the names on side a are those of places, from which personnel are assessed (A-DU) in sets of 6 workers, then DA-RI-DA and PA$_3$-NI, contribute two sets each, U-*325-ZA one set, DA-SI-*118 four sets, and the last three presumed toponymns (KU-ZU-NI, TE-KE, and DA-RE) contribute two sets combined. It is tempting to rank these presumed toponymns either in importance or in size: KU-ZU-NI, TE-KE, and DA-RE small; U-*325-ZA relatively small; DA-RI-DA and PA$_3$-NI relatively large; and DA-SI-*118 large.

- On side b, two individuals have associated with them separate logograms each also a separate unit, PA with QE-KA and KA and DI with TE-TU.

b.3: RE-DI-SE was originally written on line 3, then erased and moved to line 4d; KA & DI are therefore probably logograms functioning either as transaction signs (Schoep 2002, p. 78) or as abbreviations of types of persons associated with QE-KA and TE-TU[ ], according to the Continuity Principle, section 18..

- QE-KA appears only on HT 85b and TE-TU's two other occurrences (HT 7, HT 13) do not include these related logograms. On other tablets, KA, DI, and PA are, however, common, KA especially with personnel, perhaps women (Godart 1984, 125; cf. HT 28), DI with OLE and VIN, PA with GRA.
- Perhaps they signify separate, but conventional, functionaries specializing in being responsible for workers in certain tasks. If so, then QE-KA oversees one set of general workers and her PA oversees another tasked with, say, the harvest; and the two unnamed KA and DI functionaries associated with QE-KU might be responsible for women workers and those involved with oil or wine production. Together QE-KA and TE-TU and their functionaries were responsible for 30 workers, in 5 sets of 6 each; they would seem, therefore, be to supervisors or purveyors or collectors of personnel. Additionally, we could imagine the named individuals, QE-KA and TE-TU, associated each with a large region capable of contributing more than one set of workers and thus needing one or two assistants, such as the regions DA-SI-*118 and DA-RI-DA or PA^3-NI, already hypothesized above as "large".

***118**, syllabogram and logogram, HT 12.4, 24b.1,2,2, 38.3; KN 2.2 (TALENT?)

***120/GRA** = barley (Palmer 1995); one of "the Aegean triad" (along with *30 FIC, *122 OLIV; Schoep 2002, 98). GRA usually begins a list: GRA, OLE, *304, VIN, then either OLIV or FIC. Since GRA rarely occcurs with *303 (cf. HT 99a), they may have been handled differently (cf. HT 110a with *303, b with GRA). The largest amounts of recorded GRA: HT 15.1, 684 units, HT 40.1, 207; HT 102.1, SA-RA_2 GRA 976.

Ligatured (presumably commenting on aspects of the grain, e.g., raw, milled; destination as food or fodder; Schoep 2002, 104)

- GRA+BOSm (*576)
- GRA+DA (*573)
- GRA+KU (*579
- GRA+PA (*574), in very small quantities; so, "processed"? Compare PE 1 which allocates GRA+PA in half units to personnel (Schoep 2002, 106, 108)
- GRA+PA_3 (*577)
- GRA+QE (*578)
- GRA+QIf (*575)

When modified by fractions, it may have functioned as a land measure, as it does in Linear B (Schoep 2002, 154): *580 (+B), *581 (+E), *582 (+F), *583 (+H), *584 (+KL_2), *585 (+L_2), *586 (+L_3L_3).

***122/OLIV** = olives (Palmer 1995); one of "the Aegean triad" (along with *30 FIC, *120 GRA; Schoep 2002, 98); the total amount of olives in the HT texts are 93J units (about 8.928 liters). Olives are associated with the transaction sigh TE, but never with SA-RA_2. As in Linear B, olives and figs hardly ever appear together (exc. HT 44, 91, 131).

Ligatured

- *587 (+TU; cf. *621 OLE+TU, the only other occurrence of a sign ligatured +TU)

OLIV by itself is more frequent (HT 14.1-2, 3-5; HT 21.4; HT 44a.3; HT 50a.4; HT 58.3; HT 91.3; HT 101.6; HT 116a.3-4; HT 123+124 [5 times]; HT 131b.2; HT W 231A; ARKH 3a.2; SY Za 2a; ZA 6a.3-4, a.4; ZA a.11, b.2a; ZA 18a.2) than OLIV+TU [only at HT: 50a.4; 101.6; 116a.4]. Since OLIV+TU always appears just after OLIV, it seems to imply a contrast, and since it is rare it probably implies an uncommon contrast. In Linear B, OLIV appears 13 times at Knossos, twice at Mycenae, 8 times at Pylos. OLIV+A (agrios, wild olives) appears only at Knossos (8 times); OLIV+TI (tithasos, domesticated) appears four times at Knossos and once at Mycenae. KN E 669.1 contrasts OLIV+A 43 with OLIV+TI 45, and 669.2 mentions only OLIV, presumably the default normal OLIV. Similarly KN F 844.b lists OLIV+A 10 T 2 and repeats this information on the bottom side (label?) but only as OLIV 10 T 2. Finally, KN F 852.2 contrasts OLIV+A 70 with OLIV+TI 20.

If, in Linear A, unligatured OLIV was the domesticated OLIV, then it would be reasonable to assume that OLIV+TU was the wild olive.

It is less certain that OLE+TU (HT 23a.2; HT 28a.2-3; TY 3a.7) represents the oil from wild olives, since ligatured OLE is common and there is no pattern in the appearance of OLE+TU.

However, it should be noted that ligatured +TU occurs only with OLE and OLIV.

***123/AROM**; on Linear A clay documents this sign is a syllabogram of unknown value (A-*123-TE, DU-*123-A, TA-I-*123, TE-*123, and]A-ME-*123, all names in lists); on Hieroglyphic seals, it is a commodity

***131/VIN** = wine (Palmer 1995). Wine can be inventories in large quantities (e.g., HT 13, 103J units, or 3,744 liters; ZA 4, 104 units). Also a syllabogram (only in PU-VINa on HT 14.1, HT 123a.3-4); the sign is similar to the Egyptian sign for wine.

VIN comes in 3 variations (Schoep 2002, 100)

VINa or plain wine, flavored (as below, in ligatures)

VINb wine of lesser quality (in Linear B): ARKH 2.1-2, 3a.2, 5.3; KH 5.2 (VINb+WI); ZA 10b.1-2

VINc vinegar (in Linear B, for perfume making): KH 18.3, 4; KH 61.2; KH 85.2; perhaps PH 7b.3

Modified

- ME VINa (*507, honey flavored?), ZA 15a.3
- VINa • GRA (*597), HT 27b.6
- VINa • SI (*590), HT 27b.2

Ligatured

- VINa+KA (*596), ZA 6b.2
- VINa+RA (*594), ZA 6b.2; ZA 15b.3
- VINa+RA (*595, variant of the preceding), KE Zb 5
- VINa+SA (*589), HT 131b.2; cf. SA VINa (*522, sesame flavored?), HT 114b.1; ZA 15a.1
- VINa+SU (*593), ZA 5a.1
- VINa+TE (*588), KN Zb 34
- VINa+WA (*591), HT 27b.5
- VINa+WA (*592, variant of the preceding), HT 27b.1, 5

- VINb+WI (*598), KH 5.2

***164**, only at KH: textile?

- *164a, KH Wc 2042-4
- *164b, KH Wc 2040
- *164c, KH Wc 2041
- *164d, KH Wc 2036-9, 2045, 2111

***171**, logogram on ZA 6a with agricultural commodities (perhaps also on THE Zb 5); in Linear B, it occurs with livestock (fodder? TH nodules). This may be the same object as the "Short Column" that occurs as filler on a few seals and gold rings (Younger 1988 [Iconography, p. 280], e.g., CMS V, no. 173, at right.

***180**, PH? 31; MA 4, 6b,d, Wc 7: hide (Schoep 2002, 131, 133). On MA 4a, *180+SA contrasts with plain *180; on MA 6, the quantities are large (6a: 941, 6b 620).
Ligatured

- HIDE+[?] (*599)
- HIDE+SA+B (*600)
- HIDE+SA+L (*601)
- HIDE+B (*602)
- HIDE+L (*603)

***188**, meaning unknown, common.
Ligatured

- *188+KU (*604)

***191/GAL**, KH Wc 2028 = helmet (= Linear B *191; Palaima 1988: 325) on KH Wc 2028

***301**, common, meaning unknown.
Ligatured Schoep 2002, 113)

- *301+MI (*605)
- *301+*311 (*606)
- *301+*351 (*607)

***302**, common, olive oil (Palmer 1995), mostly in whole units; cf. TY 3 which lists olives and oil together (cf. SY Za 2).

Types of OLE in **Linear B**

- *a-re-pa*, *aleiphar*, unguent
- PA *pa-ko-we*, sage-scented
- PO *po-ni-ki-jo*, dyed red
- WE *we-ja-re-pe*, good for anointing
- SI ?
- O *origanon*, oregano-scented

Ligatures of OLE in **Linear A**

- OLE+A (*609) = A-RE-PA? (Schoep 2002, 117 n. 97)
- OLE+DI (*608), always appearing with OLE+MI (*622)
- OLE+E (*613)
- OLE+KI (*617), perhaps OLE KI-RO (owing)
- OLE+KI (*618, variant of the preceding)
- OLE+KI+U (*619)
- OLE+KI+ME (*620)
- OLE+MI (*622), always appearing with OLE+DI (*608), in whole units and small fractions
- OLE+NE (*612), in small fractions
- OLE+QE+DI (*623)
- OLE+QIf (*611)
- OLE+RA (*616)
- OLE+RI (*614), in small amounts
- OLE+TA (*615), in small fractions
- OLE+TU (*621), cf. OLIV+TU (*587, the only other occurrence of a sign ligatured +TU
- OLE+U (*610), almost always listed first and in whole units

***303**, common, a grain (Palmer 1995). It acts like Linear B *121 (wheat; Schoep 2002, 112). This commodity appears on miscellaneous tablets, usually in whole numbers (esp. 3 and 6; Schoep 2002, 101). Since GRA rarely occcurs with *303 (cf. HT 99a), they may have been handled differently (cf. HT 110a with *303, b with GRA).

Ligatured, with fractions

- *303+D (*624)
- *303+D+*304+PA (*625)
- *303+E (*626)
- *303+K (*627)

***304**, common, agricultural product, accompanying GRA on HT 92, 116 OLE and olives, GRA, VIN in large whole numbers on ZA 6 (Melena 1983, 116: sexual marker for OLIV & *303), but always found in a fixed position between OLE & OLIV. Schoep 2002, 124, suggests a spice like Linear B *kuparo* or coriander.

Ligatured

- *304+PA (*629), not found with plain *304

- *304+PA+*303+D (*630)
- *304+PA+*316+D (*631)
- *304+[ ] (*628)
- *304+[ ]+*303 (*632)
- *304+[ ]+*303+[ ] (*633)

***305**, common at HT, something to do with personnel: HT 63, with two individual entries; HT 100, with groups of VIR. It also appears in signgroups, apparently as a syllabogram

***306**, common at KH, meaning unknown. Schoep 2002, 125, suggests BOSf (it looks somewhat similar to Linear B *109b; it occurs with BOSm on KH 87; cf KH 6).

Ligatured

- *306+MI (*634)
- *306+VIR+*307 (*635)
- *306+*303+E (*636)

***307**, HT 27a.2, 32.1, 89.1, 127b.4, meaning unknown.

Ligatured

- *307+*301[(*637)
- *307+*387 (*638)

***308**, ARKH 7.3; HT 23a.1, 32.1, 35.2, 123a,b; KH 12.1, 85.1; in a word: 23-308, HG 26b.4, agricultural commodity or measured in the same way, in proportion to OLIV, always in fractional quantities, except on HT 123 (25H representing contributions?); "repeated combination of the same commodities on [miscellaneous tablets] and their consistently small quantities" suggest disbursements, not contributions (Schoep 2002, 126).

***309**, only TY 2 in three variations *309a , *309b , *309c
***312 KU**, ARKH 3a; HT 49a.6; KH 8.4 (cf. Younger 2005); for *536 (TELA+*312) Bennett 1975: 61, & Melena 1975: 108-10 both give TELA+ZO?). Younger 2005 demonstrates that Hiero sign *084 , which resembles an animal (sheep?) head, corresponds to 3 subunits of *051 , which looks like a small dagger (sheep-shearing knife?). Linear A *312 also looks like a small dagger and appears as an adjunct to Linear A *54 TELA (HT 38.3: TELA+*312; cf. TELA+KU on HT 38.3 (same line); and KU-TA[on HT 115b.4 with *312+TA on HT 10b.2).

***313** ligatured with VIR in three variations *313a (HT 89.3, 100.2), *313b (HT 94a.2; KH 7a), *313c (HT 127b.5)

***316**, agricultural commodity on KH 91.2; PH 1a.1,2, meaning unknown.
Ligatured

- *316+TO+VINa (*639)
- *316+KI (*640)

***317**, ARKH 3a.3,5; HT 96a.3, meaning unknown.
Ligatured

- *317+KU+*334 (*641)

***318**, HT 45a.3, b,3, meaning unknown.
Ligatured

- *318+[ ] (*642)

***319**, HT 132.1, Zd 1555, 156(bis), 157

***322**, KH Wc 2026, 2027, 2098; cf. Linear B *158 (unidentified) on KH Wc 2026, 2027

***323**, HT 96a.3,4

***326**, HT 35.1, 91.1, 128a.1

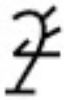
***327** = B *140 AES?, HT 97a.1, 119.1 (Palaima 1988: 326), on HT 97a.1-2, HT 119.1 where it is allotted to people. It may have had a phonetic value, since it occurs in *327-JU on HT 119.4

***330**, meaning unknown, only in ligature.
Ligatured

- *330+DA (*643)
- *330+SA (*644)

***332**, HT 97b, 107.3, agricultural product

***333**, HT Wc 3009, HT Wc 2010, MO Zf 1. Olivier 1987c, 143-44, compares the sign to Linear B *18 , for which Melena 2014, 83, suggests the value *sto*.


***334**, KH 6.6, 7a.1, meaning unknown, only at KH where it occurs first on documents and is followed by whole numbers; thus, individuals (people, livestock; Schoep 2002, 126)

***336**, KH 14.2, 82.4 = dog (based on the shape; Schoep 2002, 126-7)

***337**, only in ligature, chariot frame (cf. Linear B *242), on KH Wc 2056, 2057.
Ligatured

- *337+*188 (*645)

***338**, KH Wc 2067, 2068 = boots?

***339**, PH 12a, 15a

***341**, only in ligature, meaning unknown.
Ligatured

- *341+*188 (*646)

***343**, HT 93a.7, meaning unknown

***344**, HT 96a.4, meaning unknown

***346**, HT 122b.1, meaning unknown

***347**, KH 6.3, meaning unknown

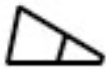
***348**, only in ligature, meaning unknown.
Ligature

- *303+*348 (*647)

***351**, only in ligature, meaning unknown
Ligature

- *351+*351 (*607)

***352** = woman MUL?, only in VIR-*352-JA, KH Wc 2100


***353**, PH 9a

***354**, PH 10

***355**, PH 10

***356**, PH 13b

***357**, PH 15b

***358**, PH 16a.2

***359**, PH 17a.1

***360**, PH 17b

***362**, ZA 10b.1: WA-*362 or TELA-*362

***365**, HT Wa 1849

***366**, HT Wa 1850

***367**, HT Wa 1851

***368**, HT Wa 1852

***369**, HT Wa 1853

***370**, HT Wa 1854

***371**, KH Wa 1018

***400-418** = vessels

- *400, handleless cup
- *401 two-handled cup

Ligatures

- *401+[ ] (*649)
- *401+A (vowel) (*650)
- *401+RU (*651)
- *401+RO (*652)
- *401+*304 (*653)
- *402, straight-sided, handleless cup
- *403, chalice
- *404, cup with one handle

Ligature

- *402+A (vowel) (*654)
- *405, two-handled bucket

Ligature

- *405+Ω (*655)
- *406, lidded bucket

Ligature

- *406+KE (*656)
- *407, lekane or basin

Ligature

- *414+A (vowel) (*657)
- *408, tripod with two vertical loop handles

- *409, handleless tripod
- *410, tripod with two vertical loop handles
- *411, tripod with two horizontal strap handles
- *412, ewer

Ligature

- *412+E (fraction) (*658)
- *413, ovoid jug (funnel?)

Ligature

- *413+SU (*660)
- *414 squat jug

Ligatures

- *414+[ ] (*661)
- *414+F (*662)
- *415, pithoid jar
- *416, pithos
- *417, wicker basket (like those the girls use in the Xeste 3 fresco for collecting crocus stamens)

Ligature

- *416+L_2 (*663)
- *418, bull head rhyton

Ligature

- *418++L_2 (*664)

Several vessel logograms have fractions written inside (see MA 10); these may refer to the capacities of the vessels or to agricultural products and their measures, since they occur in mixed commodity tablets (cf. Linear B *123 AROM which looks like a container but actually refers to the contents, or to MU-container and *sa-pi-de* boxes on PY Vn 19, MY 105; Schoep 2002, 127-8). On KN K 700 and in the Gg tablets, the numbers refer to the contents, not the numbers of the vessels (Schoep 2002, 128). The KH roundels record several vessels, according to the number of seal impressions: *408 tripod, 9; *409 tripod, 27; *411 tripod, 64; and *417 wicker baseket, 2.

12. The Libation Formula

The Libation Formula appears in part on various inscribed objects; the words, however, follow a strict sequence, which most inscribed objects adhere to but may leave out certain words (especially the three last ones) or may substitute variations (especially on the first word); the third word is always different from object to object.

word 1	word 2	word 3	word 4	word 5	word 6	word 7	word 8
T/A-TA-I-301-etc.	toponymn	person's name?	J/A-SA-SA-RA	U-NA-KA-NA-SI	I-PI-NA-MA	SI-RU-TE	I-NA-JA-PA-QA

For this sequence, B. Davis deduces (2011/2014) the word-order: word 1 is a verb, “dedicates”, word 2 is the place of dedication, word 3 (always different) must be the subject of the verb (i.e., the dedicant), and word 4 the object of the verb (“[this] dedication) — these four words constitute a primary clause. Word 5 then starts a subordinate clause with a verbal, like word 1 (e.g., “expecting [vel sim.]) with the following words the object of the verbal, i.e., what the dedicant hopes in return.

- word 1: A-TA-I-301 (etc.) has several variations (A-TA-I-301-WA-JA, A-TA-I-301-WA-E; TA-NA-I-301-TI, TA-NA-I-301-U-TI-NU, TA-NA-SU-TE[ ]-KE, TA-NA-RA-TE-U-TI-NU), all of which may argue for a phonetic value of “TSU” (vel sim.) for *301+U.
- word 2 is apparently a place-name, e.g., DI-KI-TE [undoubtedly Mt Dikte], I-DA [perhaps Mt Ida], SE-TO-I-JA [Owens 1993: Arkhanes, Mt Ioukhtas], TU-RI-SA [JGY: Tylissos?]
- word 3, always different, is probably the name of the dedicant (on PK Za 11, there are two "dedicants" PI-TE-RI and A-KO-A-NE; word 5 thus changes from U-NA-RU-KA-NA-SI [always with a single word in 3rd place] to U-NA-RU-KA-NA-TI as if to reflect a shift to the plural; similarly, A-KA-NU-ZA-TI in KN Zc 7, with 2 names in 2nd place.
- word 4: J/A-SA-SA-RA is not the name/title of a divinity, as often assumed based on its similarity to the Semitic goddess “Asherah”, but rather the word for what is dedicated, i.e., ([this] dedication”.
- if word 5 is a verbal, it would begin a subordinate clause ("dedicates," vel sim.), giving a reason for the dedication (e.g., “hoping for …”)
- word 6 has only one variation, on PK Za 10 and 11: I-PI-NA-MI-NA -- since on PK 10 word 6 follows]SI, presumably the usual ending for U-NA-RU-KA-NA-SI, the change from I-PI-NA-MA to I-PI-NA-MI-NA is not due to the change from singular to plural verb; instead, the ending may resemble the alternative ending for JA-SA-SA-RA-ME, JA-SA-SA-RA-MA-NA on KN Za 10 or JA-SA-RA-A-NA-NE on KN Zc 7.
- word 7: the -TE ending may be the same as the suffix -TE that means "of/from"

If the above inferences are correct, the Libation Formula would then translate to something like this:

“dedicates, at Place, PersonName a dedication hoping for a favor from [the divinity]”

13. Grammar

13a. Nouns, Verbs

The Libation Formula (summarized above) presents a somewhat clear dedicatory statement in a more or less strict word order in which it is possible to detect place names (word 2: e.g., Setoia, Ida, Dikte), a NAME (word 3: always different, the dedicant), and several other words, including JA-SA-SA-RA (word 4, “dedication”), U-NA-RU-KA-NA-SI (verbal/participle word 5) to I-NA-JA-PA-QA (words 5-8): “hoping for a return divine favor.”

Document PK 11, however, presents three changes in the Libation Formula:

1. instead of a single word in position 3 (NAME?), it gives two, apparently two personal names (two dedicants?)

2. word 5 changes from U-NA-RU-KA-NA-SI to U-NA-RU-KA-NA-TI, and
3. word 6 changes from I-PI-NA-MA to I-PI-NA-MI-NA.

To explain these changes, JGY hypothesizes:

1. since word 3 in the Libation Formula is always *hapax legomenon*, it is likely to be a personal name (of the dedicant?). On PK 11, the two words, instead of one, would then be the names of two dedicants.
2. If U-NA-RU-KA-NA-SI were a verbal (like a participle), then U-NA-RU-KA-NA-TI would be plural participle, corresponding to the plural subject, meaning something like "hoping."
KN Zc 7: A-KA-NU-ZA-TI • DU-RA-RE • A-ZA-RA • JA-SA-RA-A-NA-NE • WI-PI-[•]. Note the ending -TI for the first word, a variation on U-NA-KA-NA-SI/TI, followed by two NAMES before JA-SA-SA-RA
3. Similarly, I-PI-NA-MI-NA might also be a plural, perhaps a plural noun, something like "dedications."

Two transaction terms have similar endings and might also be verbs: KI-RI-SI/KI-RI-TA2 (perhaps the verbal KI-RO, "is/are lacking"), and U-MI-NA-SI ("owed"? according to the context on HT 28b.1).

13b. Adjectives

From the place name SU-KI-RI-TA, there derives a clear adjective: SU-KI-RI-TE-I-JA (HT Zb 158b), appearing on a pot that would thus be labeled "Sybritan" (cf. the similar adjective WA [for wa-na-ka-te-ro] and names in the genitive on stirrup jars inscribed in Linear B).

From this clear example of an adjective, we can identify others:

]**PU2-RE-JA** (PK Za 16) from]PU2-RA2 (ZA a.6)
KU-PA3-RI-JA (HT 24a.1) from KU-PA-RI (PE 1.1-2)
PA-SA-RI-JA (HT 24a.4) perhaps related to PA-SE-JA (HT 93a.8); compare I-PA-SA-JA (KH 10.3).

13c. Prefixes

A few words appear both in root form and with I- or J- prefixes (I+consonant, J+vowel). Duhoux 1997 identified these prefixes as meaning "to/at".

I-DA-MA-TE (AR Zf 1, Zf 2), "to" DA-MA-TE (KY Za 2)
I-ZU-RI-NI-TA (PH 6.2) related to ZU-RI-NI-MA (KN Zb 52) and perhaps A-RI-NI-TA (HT 25a.3; ZA 8.2-3)
I-PA-SA-JA (KH 10.3), "to" *PA-SA-JA; cf. PA-SE-JA (HT Wc 3001, 3002) and the adjective PA-SA-RI-JA (HT 24a.4)
JA-SA-SA-RA (common), "object case" A-SA-SA-RA (PO Zc 1)

13d. The Suffix -TE/TI

Valério 2007 demonstrated that the suffix -TE means "from/of." There is a variant, -TI.
The suffix occurs in two forms, with or without the prefix J/A-.

prefix J/A-, suffix -TE or -TI

A-DI-KI-TE-TE-DU-PU2-RE (PK Za 11), <b?ja-di-ki-te-te-du-pu2-re< b=""> (PK Za 11), (PK Za 8.a, Za 15) = "master, lord" (DU-PU2-RE) "of/from Dikte." </b?ja-di-ki-te-te-du-pu2-re<>
A-TU-RI-SI-TI (KN Zb 5), "from" TU-RI-SA (KO Za 1)

no prefix J/A- (on two interrelated documents, Hooker 1975)

DA-KU-SE-NE-TI (HT 104.1-2), "from" DA-KU-SE-NE (HT 103.4)
I-DU-TI (HT 104.2-3), "from" a hypothetical *I-DU (Ida?)
PA-DA-SU-TI (HT 104.3-4), "from" a hypothetical *PA-DA-SU
RI-RU-MA-TI (PH(?) 31), sheep and pigs "from" RI-RU-MA (HT 118.4, recording pigs, SUS I+[?]). Of this pair, Schoep 2002, 172, remarks: "It would be interesting to know whether the suffix TI indicates a direction."

14. Fractions

See Hallager 1996: 29; Pope 1960.
Linear A's fractional notation has received much attention. One aspect that remains puzzling is that there is no separate sequence for volumes/capacities and wet or dry weight. We thus have no way of knowing what the fractions refer to -- the Minoans must have switched references depending on the commodities being measured.

Bennett 1999 notes the "aliquot" system of Linear A fractions (1/x, etc.: 1/2, 1/3, 1/4, 1/5, etc.); that Egyptian Hieroglyphic 1/2 and 1/4 are the same in Hieratic, but the rest of the fractions differ in the two scripts; that Egyptian fractions are also aliquot with six signs for 1/2, 1/4, 1/8, 1/16, 1/32, 1/64 plus a monogram of 1/2+1/4 for 3/4 (the values of Linear A's fractions seem to be almost identical); and that Egyptian Hieroglyphic 1/2 is very like Linear A 'J', and 1/4 is roughly like Linear A 'E'.

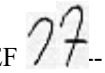
Bennett proposed correct fractional values

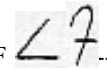
1/2 = *707/J -- correct

1/4 = *704/E -- correct

1/8 = *705/F -- correct

3/4 =*732/JE -- correct

3/8 = *721/EF  -- correct

5/8 = *735/JF  -- correct

other fractional values can be deduced

*706/H -- should be 1/6

*717/DD -- should be the double mina

*703/D -- should be the single mina, perhaps 1/5

*736/JH -- should be 2/3

Fractions in combination, in probable order of descending value

JJ (PH 9b)						
J	A (HT 120.3)					
J	E (lots; EJ [HT 123a.3-4; ZA 8.4])					
J	E	B (HT 27a.8)				
J		B (HT 129.1; KH 5.4, 6.8, 17.3)				
J			F (HT 51b.2)			
J					K (HT 32.1)	
J				H (HT 93a.3)		
J	E					L2 (KH 7a.5, 56.1)

J						L2 (HT 123b.4)
	EE (PH 12b.2, 13a, 13c)					
	E	B (KH 9.2)				
	E		F (HT 8b.4, 16.3, 40a.4, 123b.5, Zd 156)			
	E					L2 (HT 33.3)
	E					L4 (KH 26.2)
	E					L6 (KH 76.2)
	E					YYY (PH 26)
	A	BB (KH 86.2)				
		BB (KE Wc 2b)				
			F		K (PH 1b.2)	
			F			L (ZA 7b.8)
				H	K (HT 34.3)	
					K	L2 (HT 86a.2, 120.2; KH 11.2.3-4, 4.6, 16.1, 75.2)
						L2L4 (HT 33.2)
						L3L3 (HT 15.2)

Fractions with values (if demonstrable or surmisable)

A (A701). In the discussion to HT 123+124, it is clear that A[= 7/12. ABB, however, also occurs (KH 86); logically, A should be greater than B, but if it is smaller (as ABB seems to demand if B=1/3), then perhaps A is something like 1/6 (see note to HT 34.7; but see fraction H).

B (A702) = 1/3 (or 1/5). B occurs singly, in pairs BB (ZA 8, 6; KE Wc 2b), as a pair after A (KH 86.2), and once after E (1/4). That it occurs in pairs may imply that three B's would equal a unit, and that B = 1/3. The conical cup fragment PK Zb? with RE B supports this identification. Since EB occurs (KH 9.2) and JEB (HT 27), it would seem logical that B is less than 1/4 (J = 1/2; E = 1/4; see below); but on KH 9.2, EB occurs after K (1/16?), and it is therefore tempting to read this set of fractions retrograde (BEK); if so, then a descending sequence could be maintained (1/3, 1/4, 1/16). An analysis of KH 7, however, strongly suggests that B = 1/5, which would go well with B's appearance after E.

If, however, X (A711) is the half-mina and W is the mina, then W would be 1/60 talent -- that might explain the formal relationship between B (if B were 1/3) and W: W would be a conjoined BB with an implied value of 1/10 x 1/6.

D (A703) = ?1/5 (suggested by Dr Dieter Rumple; also see Double Mina, below). HT 115a.4 writes D four times, which suggests that five D's might equal a unit. It might, however, be a weight, equivalent to the mina (see below).

DD (A717) is a weight, the Double Mina (D = single mina? [see above]. The demonstration is presented in my article "Cretan Hieroglyphhic Wool Units (LANA, double mina,") Younger 2005. See below.

E (A704) = 1/4 (Pope 1960) occurs 52 times, the 2nd most common fraction (Hallager 1995); see HT 9.a

F (A705) = 1/8 (Pope 1960). On HT 8b.3-6 (list 2), the numbers total 9 + 3J + E + 2F, or 10 3/4 + 2F -- if F = 1/8, the numbers total 11. On HT 93a-b.1, the total is]165H; the numbers total 159 4J E F[H, or 161 3/4 F[H, leaving a difference of 3 F, if F is 1/8, to be filled by an entry on line a.9 or b.1.

□ H (A706) = ?1/6 (see notes to HT 123+124; also see HT 6, 94, 100); by shape related to E 1/4, D 1/5, F 1/8

□ J (A707) = 1/2 (Pope 1960), occurs 93 times, the most common fraction (Hallager 1995); see PE 1, ZA 4a.4, HT 9.b, HT 104

□ JE (A732; A707+A704; Brice 1961: 7-8, table 2) = 3/4, occurs 25 times, 3rd most common fraction (Hallager 1995)

□ K (A708) = 1/16 (Pope 1960; see the discussion to HT 155+156+157)

□ L (A709), L_2 (A709-2) (but see below), L_3 (A709-3), L_4 (A709-4), L_6 (A709-6) = values are unknown

□ L_2 (A709-2), The discussion to KH 7 suggests that L_2 = 3/20. This could mean that L is twice this, 6/20 or 3/10; and that the other L-fractions are subfractions of L: L_3 would be 1/3 L, or 6/60 or 1/10; L_4 6/80 or 3/40; L_6 1/20 -- L_5 is not extant, but it would represent 6/100 or 3/50. The L subfractions occur in combination with other fractions: EL_2 (KH 9.5, KH 13.3), KL_2 (KH 11.2, KH 16.1), EL_4 (KH 26.2, KH 75.2), and BL_6 (KH 7a.6) and EL_6 (KH 76.1).

□ W (A710) = value unknown (only at Khania: KH 12, 21.1, 60.2, 61.4, 77); the family "resemblance" between this sign and X (A711) might relate the two values (like W is 1/2X or 2X). If X is the half-mina, then W could be a full mina (1/60 talent), which might explain its appearance as two conjoined Bs (1/3 + 1/3 = 1/6 x an implied 1/10) (see below).

□ X (A711), occurs on HT 91.1 and KH 9.6 -- these occurrences do not suggest a value. The discussion to HT 123a (also HT 91.1; KH 9.6) suggests the unlikely value of 13/12. Formally, the sign looks related to 710 W ; it may represent a doubling of that ("4/3" or "1 1/3"), or a double 701 A, i.e., "1/6"?.

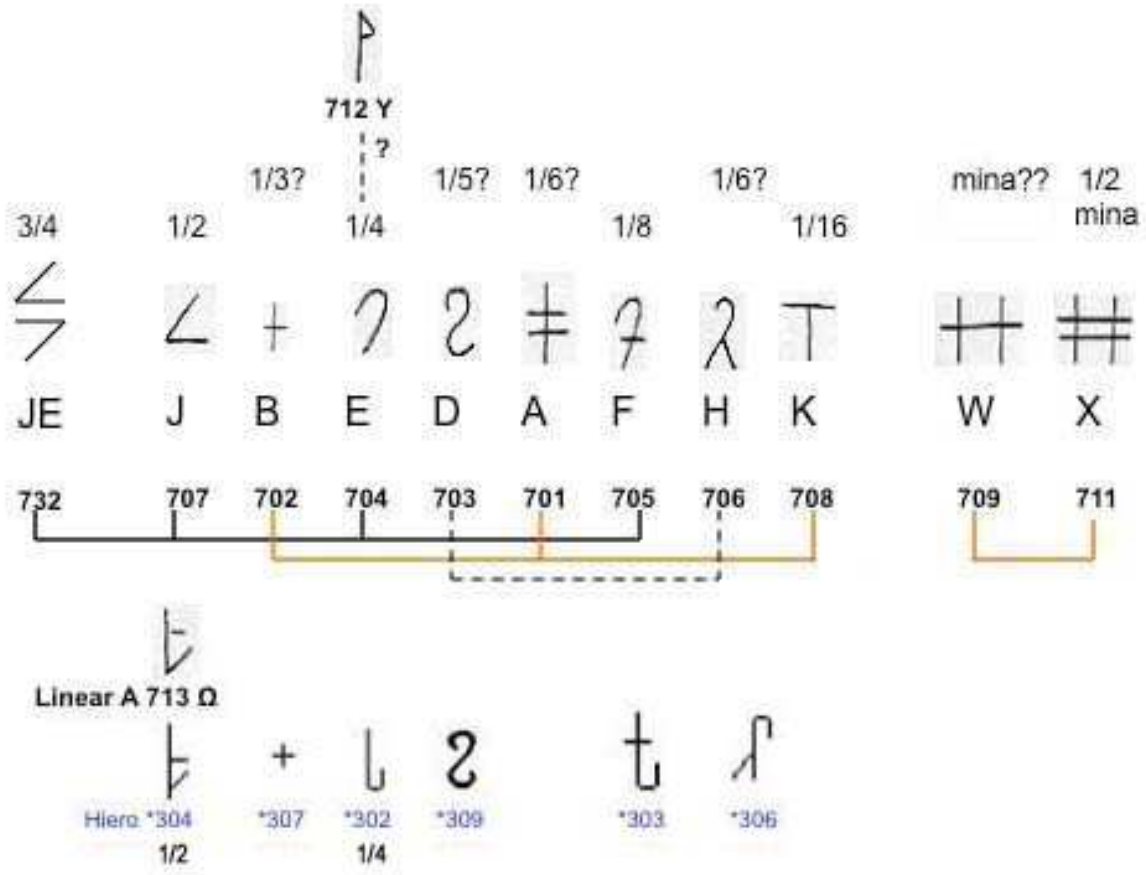
Or it may be the Linear A equivalent of Linear B *116 N, the half mina -- if so, then W could be the full mina (see above).

□ Y (A712) = value unknown, but it appears as a set of 3 (PH 26), so perhaps 1/4 (cf. PH 9a). Formally, the sign might be related to the 1/2 series.

□ Ω (A713) occurs only once, at MA (MA 10b.1), as an adjunct to A405^{VAS}. The sign is identical to Hiero *304 Λ , which can be demonstrated to be 1/2.

Summmary

For four of the fractions (J, E, F, K), we can demonstrate certain values, and can suggest, with much less certainty, values for an additional five fractions. The chart below also gives Hiero fractions that are similar in shape.



The horizontal lines below the fractions point out "family" resemblances in shape.

15. Metrology

Minoan dry unit equals the Mycenaean (96 liters) (Palaima 1994).

Liquid units: The pithos **ZA Zb 3** records "VIN 32," probably the volume, 32 units; if these are Mycenaean units (1 unit = ca. 28.8 liters), the volume of the pithos would have been 921.6 liters. Since the pithos stands about 170 cm high, its maximum capacity (as calculated from its profile by the computer program "Vase" by Gregory Christiana, copyright 1994) would have been slightly over 1000 liters. It is possible, therefore, that the Minoan unit of liquid measure was also the Mycenaean unit. See my article, "Calculating Vessel Volumes" published in Metron.

Weights

> Petruso 1992: 17-20 posits a single Minoan (and Mycenaean?) talent as "a sexigesimal multiple of its mina." Major evidence for the talent includes the porphyry triangular stone weight from Knossos (just under 29 kg; Evans 1906) and the HT bronze ingots (Parise 1967: 119: 27-32 kg). Evidence for the mina and its fractions comes from many smaller weights.

Linear A	Linear B	Denomination	Mass (gr, approximate)	Fraction of Talent	Fraction of preceding
A118	B118	talent	29,000	1/1	-
A717, DD	B117, M	double mina	967	1/30	1/30
A703, D	-	mina	483	1/60	1/2
A711, X	B116, N	1/2 mina	242	1/120	1/2
-	B115, P	1/24 mina	20.2	1/1440	1/12
-	B*21, Q	1/144 mina?	(3.36)	1/8640	1/6

Layout

16. Word Separation
Word separation is indicated in two major ways: by associating sign groups with numbers or logograms, thereby implying a separation (e.g., HT 1.2: ZU-SU **70**; HT 18.1: PA-SE GRA+QE), and by placing a dot between two sign groups, thereby explicitly separating the sign groups (e.g., HT 1.1-2: QE-RA$_2$-U • KI-RO) or between a sign group and some other sign like a transaction sign (e.g., HT 6a.1-2: DA-TA-RA • TE • FIC) or a logogram (e.g., HT 2.1: A-KA-RU • OLE+U).

In texts that employ a string of signgroups, dots separate them. This practice is most notable on non-bureaucratic texts (like the hair pins, e.g., CR(?) Zf 1) and especially in religious texts like those on the "Libation Tables" (e.g., IO Za 2).

17. "Hyphenization"
Most Linear A documents avoid splitting a word across two lines. Where a split in an expression does cut across two lines, there seems to be a semantic reason:

a sign group and a number, the most common type of separation
the components of a number (e.g., hundreds separated from units, and units from fractions)

J | E	HT 123+124a.3-4
20 | B	HT 45 b.3-4
35 = 30 | 5	HT 6b.5-6
680 = 600 | 80	HT 92 .1-2
197 = 100 90 | 7	HT 1.1-2

sign group and logogram

QIf-*118 • | FIC	KH 88.1-2
TI-MA-RU-WI-TE | *316+RO	PYR 11-2
DU-RE-ZA-SE | VINb	ZA 10b.1-2

SA-QE-RI • | KA	HT 11b.4-5
KA-U-DE-TA | VINa	HT 13.1-2

The above are obvious examples, and I think of this as a Minoan practice similar to our practice of dividing words across lines of text by placing a hyphen between syllables.

Where a split occurs within a single sign group, the reason may involve separating prefixes from base words (the root of a sign group) or base words from their suffixes. These will require a demonstration that affixes are involved.

While some examples will be apparent (e.g., U-MI- | -NA-SI, HT 28b.1-2 & HT 117a.1-2; U-NA-KA- | -NA-]SI, IO Za 21), others may require some argument. For instance, Word 5 in the Libation Formula, J/A-SA-SA-RA-ME, has already been shown to employ a prefix /I/ to signify an objective case and to cause the first syllable to change from A- to JA-.

It is also likely that A-SA is a possible prefix, occuring four times on roundels:

MA Wc <5>, a lost roundel: side a: A-SA-DA-KA; side b: GRA+E [fraction]; rim impressed once by a lentoid depicting two birds. Side a has enough room for A-SA-DA-KA to occupy a single line; its division into two lines was therefore deliberate. Since the commodity is specified (GRA), and the shipment is specified (one shipment [one seal impression]), and the supervisor is specified (i.e., the seal's "owner"), and the destination of the shipment is specifically understood to be Ayia Triada, A-SA-DA-KA must refer to the origin of the shipment. DA-KA by itelf is written on five one-hole sealings from Haghia Triada (HT Wa 1001-1005); probably therefore refers to the sealings' place of origin. The seal impression is from a gold ring (*CMS* II, 6, no. 11: two men processing). If DA-KA names the place of origin, A-SA as prefix probably characterizes it as religious – as does the gold ring.

ZA Wc 2, another roundel, presents A-TI-KA A-/-DU-KO-[[MI]] over two lines (/ indicates the break; [[MI]] was erased) and two seal impressions (*CMS* II 7, no. 25) on the rim. A-TI-KA may be name of the seal-owner responsible for two items being shipped from A-DU-KO the place of origin. The suffix -MI may have been erased because it was not needed after the prefix A-.

PYR Wc 4 presents on its obverse TI-[•]-/-I-SI written over two lines, two signs per line. The reverse presents a unique logogram which resembles Linear B AURUM (gold; Hallager 1996, II, p. 197). The roundel's rim was impressed by two different seals, one seal twice (CMS II 6, no. 234: two lions attacking a bull), the other seal three times (CMS II 6, no. 235: woman at a shrine): thus, two units of "AURUM" contributed by the "owner" of *CMS* II 6, no. 234, and three units by no. 236, all "going" (-I-SI?) to TI-[•] (Younger 2015).

GO Wc 1 presents a single word over three lines: A-SA-|-SU-MA-|-I-SE. The reverse presents the logogram BOSm **5**, and the rim was impressed five times by the same seal (CMS II 6, no. 159: bull to left), i.e., five bovines under the supervision of one person, the "owner" of the seal, "to" the religious site (A-SA) of SU-MA. If the suffix -I-SE implies the direction of shipment, "to," as suggested above, then the roundel appears to refer to the shipment of five bulls or oxen "to" A-SA-|-SU-MA under the supervision of one individual. In the fanciful piece referenced above (Younger 2015), I suggested that the two roundels from Gournia and from Pyrgos recorded the recruitment of 5 bovines from Gournia to transport 5 units of "gold" from Pyrgos to the sanctuary at (Kato) Symi.

18. The Continuity Principle

Many documents seem to have economized when listing commodities: the last cited NAME, situation, or commodity prevails until another situation is stated. I call this the "continuity principle." It is most apparent where a NAME is followed by a list of different commodities and amounts, each commodity related to that NAME until a different NAME is cited with its own list of commodities.

Two clear examples are HT 18 & 19:

HT 18

side.line	statement	logogram	number	fraction
.1	PA-SE	GRA+QE	**20**	
.1		OLE+KI	**2**	

.2		*304	**3**
.2	SA-RA$_2$	GRA	**10**
.2-3		FIC	**10**
.4-5	*vacant*		

HT 19

side.line	statement	logogram	number	fraction
.1-2	RA-*164a-TI	• TE • VINa	**30**	
.2-3	SA-RO		**5**	J
.3-4	DU-ME-DI		**43**	J

HT 18 assumes that the first 3 commodities are associated with PA-SE and the next 2 commodities are associated with SA-RA$_2$. In other words, the situation ("commodities associated with NAME") is assumed to be stable until a second name occurs which calls for a different distribution.

HT 19, however, assumes that VINa is associated with (or continues among) all three NAMES. Presumably the situation would change if a fourth name were listed with a different commodity.

HT 20 presents a more subtle pattern of continuity:

HT 20

side.line	statement	logogram	number	fraction
.1	PA-RO-SU			
.1-2	KU-MA-JU			E
.2-3	QE-KU-RE	DI		J
.4		CAPm		F
.2	SA-RE-JU			F
.2-3		WA		E
.4-5		*188+KU		J

Here, PA-RO-SU heads a short list of three NAMES. The first entry, KU-MA-JU (.1-2) could be KU-MA-JU or just KU-MA with JU being a logogram (cf. HT 97a.4). QE-KU-RE is associated with a different logogram DI, then CAPm, which is apparently the commodity also associated with (and continued for) SA-RE-JU, which is then associated with a different second and third commodity.

The documents that employ the Continuity Principle might therefore reflect short initial documents that record contributions which are then organized in an outline fashion. For instance, I can imagine 5 separate documents being collated to produce HT 20:

1	QE-KU-RE	DI	J	=	QE-KU-RE	DI	J
2	QE-KU-RE	CAPm	F	=		CAPm	F
3	SA-RE-JU	CAPm	F	=	SA-RE-JU		F
4	SA-RE-JU	WA	E	=		WA	E
5	SA-RE-JU	*188+KU	J	=		*188+KU	J

The implication of such a system is that the 5 separate short texts (really more like chits) were brought together, not because they represent the same contributions or contributions from the same place/person, but because they

represent contributions organized according to a larger principle: 1) made at the same time, or 2) made from the same region/person, or 3) organized/collected by the same Collector.

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